

WHEN SCIENCE BUILDS A PARTNER

*From the Alchemist's Workshop to the Artificial
Researcher*

A story of truth, tools, institutions, and the future of discovery

A NOTE ON SOURCES, CLAIMS, AND FUTURE SCENARIOS

This book brings together the history and philosophy of science, the sociology of knowledge, research on reproducibility, and the rapidly changing technical literature on AI-assisted and autonomous research. Recent projects and benchmarks were checked through July 2026. Peer-reviewed articles are distinguished from preprints and technical reports in the selected bibliography. The purpose is not to announce that a general artificial scientist has already arrived. It is to understand what has become technically possible, what remains fragile, and what institutions would be required for machine-generated results to deserve scientific trust.

Performance figures are included only when they clarify an argument. A high score in literature synthesis does not establish experimental autonomy. A workshop-level paper does not establish general scientific competence. A closed-loop laboratory demonstration does not guarantee industrial reliability. A model that optimizes a measurable score may still misunderstand the scientific question. Throughout the book, local capability is kept separate from scientific legitimacy.

The scenes set in 2035 and 2045 are concrete scenarios, not forecasts with fixed probabilities. They combine technologies already demonstrated - retrieval over scientific corpora, tool-using agents, code generation and execution, active learning, robotic laboratories, formal verification, provenance, and automated audit - with institutional choices that remain open. The same technical capability can support an open republic of knowledge or a concentrated market in which a few organizations control the literature, compute, laboratories, and validation services.

The term artificial intelligence is used as an umbrella for machine-learning models, large language models, symbolic systems, search procedures, agents, and robotic infrastructures. When a system is said to propose, criticize, choose, or discover, those verbs describe operational functions. They do not imply consciousness, moral intention, or legal responsibility. The book's central concern is not whether a machine has an inner life. It is whether a research process can produce claims that survive organized, technically competent doubt.

An impressive technique does not become science through its power, but through the way it can be checked by people who have no reason to believe it.

Contents

Prologue	Two Rooms, One Question	1
Chapter 1	Before Science: The World That Had to Be Deciphered	4
Chapter 2	How Truth Became Public	13
Chapter 3	The Modern Knowledge Factory	22
Chapter 4	When the Tool Began to Propose	33
Chapter 5	The Laboratory That Does Not Sleep	49
Chapter 6	Sciences We Did Not Invent	59
Chapter 7	Power, Meaning, and the New Scientism	69
Chapter 8	After Peer Review: The Anatomy of Verifiable Trust	79
Chapter 9	The Republic of Human and Artificial Researchers	89
Epilogue	Science After the Author	101
Sources	Selective Bibliography and Reading Guide	105

Numbered pages begin with the prologue.

Two Rooms, One Question

On a cold night in a European town near the end of the Middle Ages, a man sets a crucible over glowing coals and watches a metal change color. His room is narrow and smoky. The jars have no stable labels. Recipes have been copied by hand, altered by memory, and protected by symbols whose ambiguity is partly philosophical and partly commercial. The operator does not have a laboratory in our sense of the word. The precision of his instruments depends less on universal units than on the trained eye and hand. He raises the heat, waits, compares, fails, and tries again. In that room, explanation, craft, spiritual discipline, patronage, and hope are not yet separate domains. Yet something recognizable as research is happening. A procedure can succeed even when the theory offered for it is wrong.

Several centuries later, in a room with almost no people, a robotic arm transfers samples between a synthesis station and a characterization instrument. An artificial-intelligence system has searched the literature, proposed the next experiment, generated analysis code, estimated uncertainty, and selected a region of the design space that seems worth exploring. Sensors stream data into an experiment registry. A second model looks for anomalies. A third checks whether the protocol has drifted from its approved version. The laboratory can work through the night. It does not become tired, forget to save a file, or lose patience after the hundredth failed run. It can also pursue a poorly chosen metric with flawless persistence. It can turn calibration drift into a discovery and produce, at the same speed, both knowledge and the appearance of knowledge.

Between these two rooms lies the history of science. It is not a straight road from superstition to truth. It is the history of societies learning, slowly and conflictually, to convert curiosity into a public practice, measurement into a shared convention, doubt into an institution, and a result into a claim for which somebody must answer. Philosophy asked what it means to know. Alchemy joined the workshop to cosmology. Printing made texts comparable at scale. Learned

societies created rituals of testimony. Instruments extended the senses; statistics disciplined the human appetite for patterns; universities, journals, national laboratories, grants, ethics committees, and peer review gave inquiry a social body. Every gain in rigor, however, created a new surface on which rigor could be imitated.

Artificial intelligence enters this history as more than one additional instrument. It can occupy several roles at once: librarian, programmer, statistician, critic, experimental designer, laboratory operator, and manuscript writer. Systems such as OpenScholar and PaperQA2 search large scientific corpora and return evidence-linked syntheses. The AI co-scientist uses a multi-agent process to generate, debate, rank, and evolve hypotheses. ERA combines language models with tree search to write and improve empirical scientific software when a task can be scored. The AI Scientist and Agent Laboratory connect idea generation, coding, experimentation, and paper production. Coscientist, ChemCrow, A-Lab, ChemOS, and related self-driving laboratories connect digital reasoning to physical instruments. Benchmarks such as ScienceAgentBench, MLE-bench, PaperBench, MLR-Bench, AutoResearchBench, and ResearchClawBench ask harder questions: can an agent write valid scientific code, find the right literature, replicate a paper, or rediscover a result under conditions designed to expose fabricated success? Their answer, so far, is not a simple yes. Performance is improving quickly, but robust autonomy remains much weaker than fluent demonstrations suggest (Chen et al., 2024; Chan et al., 2024; Starace et al., 2025; Xiong et al., 2026).

The question of this book is therefore broader than whether AI will replace scientists. The more consequential question is what science will become when searching, comparing, experimenting, and arguing are distributed across people and artificial systems. What made a scientific result legitimate in different historical periods? What counts as validation when no individual can mentally follow every transformation between raw data and conclusion? Can there be a science that is predictively and interventionally reliable but whose concepts are not intelligible to humans? What happens to peer review when the number of manuscripts, hypotheses, simulations, and experiments grows by orders of magnitude? Who will be paid and honored for the slow work of checking? How will society distinguish a genuine researcher from a charlatan when both can generate polished graphs, equations, references, and a voice of serene certainty?

The argument developed here is that the future of science depends less on the arrival of an artificial genius than on our ability to rebuild the institutions of knowledge around verifiability. Science did not become credible because scientists stopped making mistakes. It became credible because, at its best, it made error detectable, disagreement permissible, and correction possible. Artificial partners will amplify research only when important claims arrive with their paths attached: sources, data, code, versions, instruments, uncertainties, rejected alternatives, negative results, and independent attempts at refutation. Without that path, AI will not create an age of knowledge. It will create an age of industrialized plausibility.

The two rooms share another feature. In each, outsiders must decide whether to trust the operator. In the first room, the operator is a person who invokes tradition, experience, and perhaps the name of a patron. In the second, the operator is a network of models, services, databases, instruments, and organizations, each carrying a different kind of technical authority. In both cases, the local marvel may be genuine, the mistake may be sincere, and fraud may borrow the vocabulary of competence. The fundamental question is not what impresses us. It is what can be traced, repeated, and contested.

This is not a story of an inevitable future. The same capabilities can strengthen open science or deepen concentration. They can relieve researchers of repetitive labor or turn them into supervisors of systems they no longer understand. They can make negative results visible or produce an ocean of synthetic papers that hides them. They can democratize access to expertise or place scientific sovereignty in the hands of a few firms and states. Technology opens a range of futures; institutions select among them.

The narrative will therefore move repeatedly between beginnings and concrete futures. The crucible will stand beside the autonomous laboratory. Boyle's witnesses will stand beside audit agents. The journal article will be compared with an executable research object. The old patron will reappear as the owner of compute, data, and cloud laboratories. The alchemist's secret will return as a proprietary model. By placing old and new forms of authority next to each other, we can see that the coming transformation is not only about faster tools. It is about the conditions under which a civilization agrees to call something knowledge.

Before Science: The World That Had to Be Deciphered

1.1 The field, the temple, and the workshop

At the beginning, knowledge did not live in a single institution. It was distributed among field, temple, workshop, court, ship, kitchen, and story. A farmer could smell rain in the soil without possessing a theory of atmospheric circulation. A navigator could convert a constellation into a direction. A healer carried recipes whose effects mixed observation, tradition, placebo, and luck. A priest connected the order of the sky to the order of the city. A craftsperson read temperature in the color of a flame rather than in degrees. These were not merely primitive versions of modern science. They belonged to social worlds in which truth, usefulness, authority, and meaning had not yet been separated into distinct professions.

When modern readers open an ancient text, they are tempted to divide every sentence into correct and incorrect, as though its author had been sitting an examination in a discipline that would not exist for centuries. People do not build knowledge backward from our future. They work with the concepts, instruments, and institutions available to them. A cosmology could simultaneously explain the motion of bodies, justify political hierarchy, and provide a moral map. A treatise on metals could speak of sulfur, mercury, maturation, and perfection without feeling that it had crossed an illicit boundary from chemistry into metaphysics. The sharp policing of domains is itself a historical achievement, not a natural starting point.

Greek philosophy introduced a decisive ambition: the world could be explained through principles open to public argument, not only through divine genealogies. The transition from mythos to logos was neither complete nor uniform, but it created a new kind of demand. It was no longer enough to say that things happened; one should give reasons and expose those reasons to confrontation. Aristotle organized observations, classifications, and causes into an intellectual architecture of enormous reach. Yet the prestige of demonstration and

the stability of that architecture could also constrain what was asked of experience. Modern science was not born by replacing reason with experiment. It emerged by repeatedly renegotiating the relation between them.

The history was never exclusively European. Across the medieval Islamic world, astronomy, optics, medicine, mathematics, instrumentation, and translation became parts of extensive intellectual networks. Ibn al-Haytham linked the study of vision to controlled observation and geometrical analysis. Observatories accumulated systematic records; physicians combined and criticized multiple traditions; translators moved concepts among Greek, Syriac, Arabic, Persian, Hebrew, and Latin. European universities later inherited and transformed these materials through scholastic disputation, which itself developed rigorous techniques of objection and reply. Science grew through circulation, appropriation, translation, and institutional recombination. Centers often forgot the peripheries from which their tools and questions had arrived.

Alchemy is useful precisely because it resists the easy label of failed science. For a long time, it was treated as a museum of delusions from which chemistry escaped. Historians such as William Newman and Lawrence Principe have shown a more complicated world. Medieval and early-modern alchemists joined theories of matter to repeated operations of distillation, separation, alloying, sublimation, assaying, and early quantitative control. Some recipes can be reconstructed; some apparatus and techniques became part of later laboratory chemistry. Alchemy was not simply irrational. It was a mixed culture in which skilled material practice, cosmology, secrecy, medicine, craft, and patronage were bound together (Newman, 2011; Principe, 2013).

The alchemist's workshop nevertheless had an institutional weakness. Knowledge was often encoded, guarded, and attached to the reputation of a master. A recipe was not yet a standardized protocol addressed to an open community of challengers. Patronage rewarded spectacular promises, while symbolic flexibility allowed failure to be reinterpreted. If transmutation failed, the material might have been impure, the operator unworthy, the timing astrologically wrong, or the spiritual preparation incomplete. Within such a framework, a theory could absorb almost any result.

That flexibility did not disappear when white coats appeared. It returns whenever an explanatory system is protected by moving

standards. When an AI system produces a wrong prediction, its defender may say that the prompt was poor, the corpus incomplete, the tool unsuitable, the benchmark contaminated, or the evaluator unable to recognize novelty. Any of those explanations may be correct. The danger begins when they are invented after the test and no prior condition would count as failure. A future workshop may contain robots, calibrated sensors, and foundation models, yet preserve the oldest structure of secrecy: an opaque procedure interpreted by initiates and justified retrospectively.

What we now call science arose through a slow reorganization of this structure. It required not only better experiments but experiments witnessed by others; not only results but descriptions designed to travel; not only personal authority but procedures through which authority could be challenged. Earlier knowledge could be exact, profound, and useful, yet its legitimacy often remained tied to person, lineage, guild, or status. AI forces the same question at a new scale: how can the value of a claim be separated from the prestige of the system that generated it?

Measurement supplies a second thread. The meter, second, kilogram, pH scale, and genomic reference sequence can feel like natural features of the world. They are better understood as agreements embodied in artifacts, instruments, protocols, and institutions. Trade, taxation, navigation, war, engineering, and empire all created pressure for standardization. A balance that gave comparable readings in different towns was a cognitive advance and a fiscal technology. A more accurate map served astronomy and conquest. Modern science was shaped by curiosity and power at the same time.

This ambivalence will return in an age of dense sensing. An AI agent can correlate satellite images, laboratory measurements, electronic records, ecological signals, and social data at a resolution no human team could manually inspect. Yet what is not measured may vanish from its objective function. Communities without infrastructure, languages with poor digital representation, negative results that were never published, and qualities resistant to quantification can disappear from the research map. The future may suffer less from a shortage of measurements than from the belief that measurement density is the same thing as reality.

How scientific legitimacy changed

The centre of trust shifted; older forms never disappeared.



Figure 1. A compressed history of scientific legitimacy, from the master's authority to auditable autonomy.

The old craftsman also knew more than could be stated. Heat was read in the behavior of a flame, viscosity in the resistance of a spoon, material quality in sound and smell. Michael Polanyi described such competence as tacit knowledge: we know more than we can tell. Research automation must either capture parts of this knowledge through multimodal sensing, demonstration, and interaction, or recognize that competence remains embodied. A laboratory does not become complete merely because it has more data. It becomes more honest when it records what kinds of experience it has excluded.

1.2 Philosophy as a technology of doubt

Philosophy did not hand science one universal method. It offered something more durable: a vocabulary with which inquiry could interrogate itself. What is a cause? What counts as evidence? Is knowledge a justified true belief, an ability to intervene, an explanation, a reliable prediction, or a stable network of

inferences? Are objects simply given, or do instruments and concepts participate in constituting the phenomena we observe? How much uncertainty is compatible with the declaration that we know? These are not ornaments added after discovery. They shape what is measured, what is ignored, and how a result is distinguished from an accident.

Francis Bacon became the emblem of induction from observations to laws. René Descartes stood for analysis, order, and rational clarity. Galileo joined experiment, instrumentation, and mathematization. Robert Boyle turned the experiment into a public scene. None invented the scientific method alone, and their real practices were far messier than textbook portraits. Their importance lies in the new relations they articulated among experience, authority, intervention, and demonstration. Nature was no longer only contemplated. It could be arranged so that competing possibilities produced different answers.

Intervention matters because passive observation can fit many explanations. An experiment changes conditions and studies the difference. In contemporary terms, it tries to distinguish correlation from causation. Yet an experiment never speaks entirely for itself. Instruments are calibrated, samples selected, variables defined, noise modeled, and outcomes interpreted. Pierre Duhem and W. V. O. Quine, in different forms, emphasized that hypotheses are tested together with auxiliary assumptions. When a prediction fails, it may be the theory, the instrument, the boundary condition, the implementation, or the data treatment that has broken. This does not make science arbitrary. It explains why evaluation cannot be reduced to a mechanical yes-or-no rule.

Karl Popper made falsifiability a powerful ideal: a scientific theory must risk being wrong. Thomas Kuhn showed that researchers work within paradigms that define legitimate problems, trusted instruments, exemplary solutions, and ordinary standards of success. Imre Lakatos tried to preserve rational criticism by speaking of progressive and degenerating research programmes. Paul Feyerabend warned that the history of discovery does not obey a single clean method. Contemporary philosophy therefore tends toward methodological pluralism. Different sciences earn trust differently, yet they share families of norms: explicit argument, comparison, traceability, calibration, criticism, and correction (Popper, 1959; Kuhn, 1962; Lakatos, 1970; Feyerabend, 1975).

This pluralism changes the question of AI. Asking whether an artificial system can do science is too coarse. Science is not one indivisible act. It includes choosing questions, constructing concepts, designing measurements, retrieving evidence, generating hypotheses, selecting experiments, executing procedures, analyzing data, producing explanations, communicating claims, checking results, and accepting consequences. These components will be automated at different speeds. A system may be extraordinary at searching a program space and poor at deciding whether the resulting improvement matters. It may be a competent experimenter and a dangerous interpreter. It may reduce one statistical bias while amplifying a political one.

Philosophy thus becomes a technology for controlling autonomy. It forces designers to declare the kind of success being pursued. Is the goal predictive accuracy, causal identification, mechanistic understanding, clinical benefit, mathematical proof, economic value, or public justice? A score is not a natural property of the world. It is a chosen compression of what matters. When a system optimizes a metric, it does not merely discover how to reach truth. It inherits a partial definition of truth. In human research, that definition is often tacit and negotiable. In automated research, it becomes executable code and can be repeated millions of times.

The most useful philosophical work for AI-enabled science may therefore concern boundaries rather than machine consciousness. Which assumptions are active? What would count as failure? Which populations and contexts support generalization? Which values entered the objective? Which part of a conclusion was measured, which was inferred, and which was interpreted? An AI system can answer these questions only if its architecture preserves the history of its own process. Philosophy without infrastructure remains commentary. Infrastructure without philosophy becomes prejudice automated at scale.

Statistics added another discipline of doubt in the nineteenth and twentieth centuries. Variation became not merely an imperfection but an object of study. Samples, distributions, errors, intervals, priors, and likelihoods made it possible to reason about populations and processes that no one could observe in full. Yet statistics introduced conventions and judgment at every stage: the model of error, the stopping rule, the prior, the significance threshold, the handling of missing data, and the

choice of comparison. Numbers did not eliminate discretion. They relocated it into design.

This relocation is decisive for artificial research agents. A system can execute a thousand tests correctly and reach a deeply misleading conclusion because the question, sample, counterfactual, or comparison was wrong. It can estimate uncertainty conditional on one model while ignoring uncertainty over model choice. It can adapt experiments in real time and blur the border between exploration and confirmation. As research becomes more adaptive, methodological philosophy will have to become executable governance: what must be registered before data are seen, when a hypothesis may be revised, which branches of analysis must be reported, and how selection among thousands of failed paths should affect confidence in the surviving one.

There is also an ethical resemblance between epistemic virtues and character. Precision, honest reporting, willingness to publish negative results, and courage to abandon a favored idea are cultivated social practices. A model does not possess virtue in the human moral sense, but an architecture can encourage or suppress operational equivalents. It can preserve every attempt or only the winner. It can surface contradictions or smooth them away. It can be rewarded for calibration or for theatrical confidence. What we call character in a person becomes, in a system, a policy of optimization, memory, permission, and audit.

1.3 The alchemy of the future

Alchemy offers an uncomfortable comparison for the age of large models. The comparison is not that AI is magic. It is that both cases separate practical effectiveness from explanatory transparency. An alchemist could produce a pigment, alloy, solvent, or medicinal compound through a theory of matter that modern chemistry rejects. A neural model can classify images, predict structures, or propose molecules without supplying concepts that align with human explanations. In both cases, local success may be real while its epistemic meaning remains unsettled.

One response is pragmatic triumphalism: if it works, that is enough. In many applications, performance is indeed central. A stronger material, a more accurate diagnostic test, or a faster algorithm has value even when the route to it is not fully intelligible. But science does

not seek only isolated successes. It looks for relations that transfer, explanations that support new interventions, and knowledge that can be criticized by communities other than the one that produced it. A procedure that works in one laboratory, on one data distribution, under one fragile calibration does not have the same standing as a relation that survives changes of instrument, population, implementation, and context.

Paul Humphreys used the idea of epistemic opacity to describe computations containing more operations than any person can inspect in practice. Complex simulations and learned models intensify this condition. Opacity does not automatically destroy knowledge. A community can have good reason to trust a result even when no individual follows every operation. Trust must then move toward the architecture surrounding the computation: tests of code, sensitivity analyses, external validation, independent implementations, monitoring for drift, and preserved provenance (Humphreys, 2004).

This marks the difference between old alchemy and a possible disciplined alchemy of the future. The former could hide failure inside secrecy and symbolism. The latter may accept strange representations only when they undergo severe tests. A concept invented by an AI need not resemble a human concept in order to be scientific. It must, however, support novel predictions, permit discriminating interventions, survive changes in instrumentation, and remain connected to a verifiable path. An incomprehensible but testable theory differs fundamentally from an incomprehensible assertion that asks to be trusted by authority.

We are likely to encounter at least three levels of human access to machine-generated science. At the first, people understand the mechanism and can explain it in ordinary disciplinary language. At the second, people understand global properties, domains of validity, and failure conditions, much as engineers use models too complex to inspect line by line. At the third, artificial systems work with representations that cannot be translated without major loss, while humans gain access mainly through consequences, counterexamples, and verification certificates. Treating these levels as equivalent would be reckless. Rejecting the latter two in advance could also block genuine knowledge.

The issue becomes political when opaque results affect bodies, ecosystems, or the distribution of resources. A model may recommend

a treatment, but the patient is more than a point in a loss function. An agent may discover a carbon-capture process, but its location, environmental risk, and ownership are value-laden choices. A network of AIs may optimize agriculture while defining yield in a way that ignores biodiversity, farmer autonomy, soil resilience, or cultural continuity. The stronger the technical system becomes, the more dangerous it is to confuse empirical validity with total legitimacy.

Opacity also has an economic dimension. The alchemist guarded a recipe to retain patronage; a modern firm may protect a model, dataset, or laboratory platform for competitive advantage. Trade secrecy does not make a result false, but it narrows the community able to challenge it. When a public claim depends on an inaccessible system, society must decide what substitute for openness is adequate: confidential audit, access for accredited laboratories, replication through a different model, disclosure of evaluation artifacts, or cryptographic attestations that establish which code and data were used without revealing them completely.

Here explanation and traceability must be separated. We may be unable to translate the full internal representation of a model, yet still require its version, inputs, tools, permissions, calibration records, execution history, and adverse-test results. A psychological story about how a model supposedly thought may be impossible or misleading. A technical history of what it did is attainable. For scientific trust, that history may matter more than fluent simulated reasoning.

Alchemy was not overcome only by better theories. It was absorbed and transformed by a culture that changed who could see, repeat, and contest. Autonomous research will require an analogous transformation. Results will need to arrive with the social and technical conditions of their own challenge. The secret manuscript must become an executable research object. The master's authority must give way to independent evidence. The ambiguous formula must be replaced by versions, metadata, limitations, and tests. Instead of saying that the system knows, we will need an architecture showing what was observed, what was inferred, what was optimized, and what remains unknown.

The first lesson of history is therefore unexpectedly modern. Power does not make a technique scientific. The decisive question is whether those who have no reason to believe it can nevertheless test it.

How Truth Became Public

2.1 The experiment as a controlled spectacle

In the seventeenth century, an air pump could be more than an apparatus. It could be a stage on which a community learned how to believe in a fact. Robert Boyle and his collaborators did not simply introduce a device that produced a partial vacuum. They helped invent a social form of experimentation. Witnesses watched, descriptions circulated, objections were recorded, and credibility depended on whether competent persons could agree that an event had occurred. Steven Shapin and Simon Schaffer showed that the dispute over the vacuum between Boyle and Thomas Hobbes was also a dispute about political order: who had the right to declare a fact, and what kind of community could manage disagreement without collapsing into coercion (Shapin and Schaffer, 1985).

The move from secrecy to testimony was neither immediate nor democratic. Experiments were expensive, apparatus was rare, and literal replication was difficult. Public did not mean universally accessible. It meant that a result had entered a network of recognized persons, texts, instruments, and institutions. The scientific periodical widened that network. *Philosophical Transactions*, first published in 1665, became a technology for circulating observations. Experimental writing cultivated a cautious, detailed style capable of producing what Shapin called a virtual witness: a reader who had not been present could nevertheless imagine the procedure well enough to judge the claim.

Printing changed the economy of error. An oral statement could shift without leaving a trace. A printed statement became fixed, citable, and vulnerable to comparison. Fixity did not guarantee truth. It created a public memory of disagreement. Modern science grew not only by accumulating facts but by stabilizing objects of dispute. A table, drawing, description of an instrument, or sequence of measurements could travel and be examined by people who had never entered the original room.

Trust, however, remained personal. Who counted as a credible witness? In early-modern Europe, social rank, reputation, and apparent material independence mattered. The ideal of the disinterested gentleman offered an imperfect solution: a person who seemed less dependent on trade or patronage was thought less likely to deceive. From the present, the exclusion is obvious. Women, artisans, technicians, servants, Indigenous informants, colonial subjects, and enslaved people often contributed materially to knowledge while being denied equal testimonial authority. The universal voice of science was formulated inside institutions that were not universal.

The contradiction has a digital successor. The corpus visible to a research agent is not the world. It is the fraction of the world that has been published, digitized, indexed, licensed, translated, linked by metadata, and ranked as relevant. Dominant languages, well-funded journals, fields with clean databases, and papers available as machine-readable text become easier to retrieve. Negative results, local knowledge, commercial data, older literature, and research from peripheral institutions can disappear. When an agent summarizes the consensus, it first summarizes an architecture of visibility.

Modern scientific retrieval systems are beginning to reveal both the promise and the danger. OpenScholar retrieves passages from tens of millions of open-access papers and uses reranking, citation-aware generation, and iterative feedback to produce evidence-linked syntheses. Its strong performance on ScholarQABench shows that specialized retrieval can substantially improve coverage and citation accuracy over a general language model. The same study also shows why retrieval is not a magic solvent: answers can remain factually imperfect, open corpora omit licensed literature, evaluation depends on expert judgment, and a longer, better-organized synthesis can be preferred even when subtle errors remain (Asai et al., 2026). PaperQA2 and other agentic systems similarly demonstrate that iterative search, citation-graph navigation, and contradiction detection can outperform simple one-shot question answering on selected tasks. Yet every retrieval system inherits boundaries from its corpus, ranking function, query expansion, and access rights.

In the future, a public experiment may no longer be an event watched directly by people. It may be a sequence distributed across simulators, cloud services, models, robots, and databases. The virtual witness will be partly artificial. Instead of merely reading a literary

description, a validator may rerun a container, verify the hash of a dataset, compare instrument telemetry with the approved protocol, regenerate a figure, and test whether a citation supports the sentence attached to it. Publicity will no longer mean only that the article can be downloaded. It will mean that the structure on which the claim depends can be located, interpreted, and, at the appropriate level, re-executed.

This change is necessary because prose has become cheap. A language model can produce in minutes the surface form that once required weeks of drafting. It can imitate caution, technical density, rhetorical balance, and the conventions of a field. When legitimacy remains attached to the appearance of a manuscript, the cost of seeming scientific falls toward zero. The answer is not to prohibit generated prose. It is to move the center of trust toward what prose points to: observations, protocols, analysis, negative results, and provenance.

Boyle needed witnesses to separate the event from its report. We will need human and algorithmic witnesses to separate fluent narration from the process that produced it. Those witnesses must be genuinely independent. A system does not become credible because one simulated role confirms another role running on the same underlying model, data, and infrastructure. Independence is not a change of prompt. It is a difference in evidence, implementation, institutional interest, and failure mode.

Scientific publicity has always been a technical accomplishment, not a moral intention. For a fact to circulate, paper, figures, nomenclature, instruments, and postal systems had to be stabilized. In the digital age, uploading a PDF is the weakest version of openness. A dataset without a dictionary, a repository without a reproducible environment, a model without a version, or an experiment log without time synchronization is formally visible but practically opaque.

The algorithmic witness of the future will therefore be plural. One verifier may check file integrity; another may rerun code; another may search for data leakage; another may test statistical assumptions; another may inspect whether citations support claims; another may compare the result with external datasets. None is a universal judge. Each confirms a narrow property and states what it did not test. This is the most realistic promise of AI in evaluation: not an oracle, but a population of specialized witnesses whose work can itself be audited.

Such a system must include the right to contest the witness. When only the platform owner can run the validators, publicity becomes theater. Communities need portable formats, exportable logs, inspectable rubrics, and the ability to substitute independent checkers. The experimental fact gained stability through the multiplication of witnesses. The computational fact will gain stability through the multiplication of implementations.

2.2 From reputation to procedure

Peer review is often presented as though it had existed in its modern form since the birth of science. Its history is less tidy. Learned societies and journals used committees, correspondence, and editorial judgment from the early modern period, but those practices did not yet amount to the anonymous, specialized, standardized review familiar today. Historical work by Moxham, Fyfe, and colleagues shows that the term peer review stabilized only in the second half of the twentieth century and that editorial assessment long served several purposes at once: protecting a society's reputation, managing printing costs, selecting communications, and deciding what merited discussion (Moxham and Fyfe, 2018; Fyfe et al., 2022).

This matters because peer review is not a law of nature. It is an institutional answer to a problem of trust. In a small community, reputation and correspondence could perform much of the filtering. As universities expanded, disciplines professionalized, and publication volumes increased, evaluation became more formal. Manuscripts moved to specialists; editors mediated; acceptance became a signal not only of communicability but of merit for hiring, grants, promotion, and status.

Modern peer review does some things reasonably well and many things imperfectly. It can identify obvious errors, request clarification, improve reporting, and ask whether a paper is connected to relevant literature. It can assess whether the argument makes sense within a field. It rarely replicates an experiment, fully audits code, checks every reference, or explores all analytical alternatives. Reviewers work with limited time and incomplete information. Agreement can be low. Conventional ideas may receive a familiarity advantage, while prestige, language, institution, and social identity can shape judgment.

Publication does not mean a result is true. It means the work passed a social and technical filter of variable intensity.

Robert K. Merton described the ethos of science through norms that remain influential: communalism of knowledge, universalism of criteria, disinterestedness, and organized skepticism. These were never perfect descriptions of actual behavior. They were standards by which behavior could be criticized (Merton, 1942/1973). When data are hidden, communalism supplies a language of objection. When status substitutes for argument, universalism makes criticism possible. When funding produces conflicts, disinterestedness becomes a norm against which disclosure and design can be judged. When consensus hardens, organized skepticism legitimizes the dissenter.

AI can strengthen or corrode each norm. It can translate across fields, search neglected literatures, and make technical expertise more accessible, approaching a version of communalism. It can apply the same formal checks to every manuscript, approximating universalism. It can also centralize knowledge inside private platforms, reproduce the biases of its training corpus, and apply uniform rubrics to contexts that require different standards. A reviewer model trained on historical decisions can turn yesterday's conservatism into tomorrow's automatic policy.

Scientific legitimacy gradually shifted from person to procedure. We do not trust a measurement only because a famous researcher made it; we ask for calibration, controls, uncertainty, and analysis. We do not accept a drug because a clinician tells persuasive stories; we require controlled trials, safety evidence, reporting standards, and evaluation of benefit against harm. We do not accept a mathematical proof solely because of the author's prestige; others inspect the steps, and increasingly, formal systems can check parts of the derivation. Yet procedure can itself become a fetish. A checklist completed mechanically does not guarantee quality. A p-value below a conventional threshold does not transform an association into truth. A benchmark win does not establish relevance outside the tested distribution.

The future procedure must therefore be layered and proportional to risk. A low-stakes exploratory claim may pass rapid automated checks. A clinical claim, a biosafety-sensitive experiment, or a result capable of guiding public policy will require independent audit, external validation, and explicit human responsibility. A mathematical result

may receive a formal certificate while its importance remains a communal judgment. A material may be reproduced in robotic laboratories using different suppliers and instruments. Instead of a single stamp reading peer reviewed, claims may carry changing states: proposed, computationally checked, internally reproduced, independently replicated, externally validated, disputed, corrected, or withdrawn.

This is not a weakening of publication. It is a recognition that the article has always been a compressed interface to a richer process. When research becomes executable and results multiply, the compression is no longer sufficient. Trust must become updateable. A claim can gain confidence through replication and lose it through audit. The scientific record begins to look less like a library of finished monuments and more like a network of claims whose evidential dossiers evolve.

AI can make procedure more transparent or more ritualized. An editorial system can automatically test data availability, statistical consistency, citation support, image manipulation, reporting guidelines, and reproducibility scripts. It can also compress all this into a proprietary credibility score whose weights no one may challenge. Numerical precision can hide arbitrary design. A legitimate procedure is not the one that outputs the most exact number. It is the one that exposes reasons, limitations, conflicts, and paths of appeal.

Reputation will not disappear. It will fragment. A laboratory may be trusted for calibration but criticized for selective reporting. An agent may be reliable at checking code and weak at judging novelty. A data repository may have strong provenance and poor population coverage. Mature institutions will ask not whether an actor is trustworthy in general, but trustworthy for which property, in which domain, over which version, and on the basis of what record. Trust becomes granular, revocable, and scoped.

2.3 Many sciences, many regimes of evidence

The phrase scientifically proven hides the deep diversity of scientific practice. In mathematics, legitimacy centers on proof, although discovery may be guided by examples, computation, diagrams, and intuition. In particle physics, the convention of five standard deviations contributes to the announcement of a

discovery, yet the claim also depends on detector calibration, background modeling, systematic uncertainty, collaboration, and independent confirmation. In clinical medicine, randomized trials, meta-analysis, risk-of-bias assessment, adverse-event reporting, and frameworks such as GRADE organize evidence. In geology, ecology, evolutionary biology, and paleontology, many events cannot be repeated; credibility grows from the convergence of independent traces, mechanisms, models, and predictions. Climate science combines observations, paleoclimate records, physical theory, model ensembles, and calibrated uncertainty. Computer science mixes proof, benchmark performance, executable artifacts, ablations, and replication under different software and hardware conditions. Historical research depends on archival provenance, contextual interpretation, comparison among independent traces, and explicit argument about absence. Ethnographic and qualitative inquiry add sustained observation, reflexivity, triangulation, participant meaning, and a record of how the researcher's position shaped access and interpretation.

Table 1. How representative sciences authorize a claim.

Field	What grants legitimacy
Mathematics	Proof checked step by step; significance judged by the community.
Particle physics	Statistical threshold, calibration, systematics, collaboration, and independent confirmation.
Clinical medicine	Controlled trials, safety, bias assessment, evidence grading, and patient relevance.
Earth and life sciences	Convergence of traces, mechanisms, observations, models, and transportability.
Computer science and ML	Proof or benchmark, executable artifact, data provenance, ablations, and external replication.
Historical and qualitative inquiry	Archival provenance, contextual interpretation, reflexivity, triangulation, and participant meaning.

Diversity does not mean that anything goes. It means that standards must fit their objects. A randomized controlled trial may be appropriate for a drug and absurd for the formation of a galaxy. A formal proof can certify a property of an algorithm but cannot establish that a dataset represents a population. Exact replication may be crucial in chemistry and impossible for a unique earthquake; in the latter case, researchers seek conceptual replication and convergence among independent sources.

AI meets this plurality through a deceptively uniform interface. The user asks for evidence and receives a coherent answer, although the meaning of evidence differs by field. A general agent may import statistical norms from one discipline into another, mistake a preprint for clinical consensus, treat a simulation as an observation, or rank a benchmark improvement above causal understanding. Retrieval-augmented generation reduces some citation errors, but it does not automatically understand evidence hierarchies. Finding a relevant paper is not the same as evaluating its design, statistical power, population, bias, and relation to the rest of the field.

Contemporary legitimacy can be described through four linked questions. The ontological question asks what kind of object is being studied and through which operations it becomes observable. The methodological question asks which design supports the inference. The social question asks who checked the result, with what independence, competence, and conflicts of interest. The temporal question asks how the claim behaved after publication: in replications, applications, failures, and new contexts. A competent scientific agent must represent all four levels, not merely match passages of text.

Regimes of evidence may become more explicitly encoded. A biomedical agent could refuse to equate an *in vitro* effect with clinical efficacy and represent the transitions through toxicity, animal models where justified, human safety, trial phases, and post-market monitoring. A materials agent could require independent characterization and stability tests. A social-science agent could inspect preregistration, construct validity, measurement invariance, and cultural generalization. A mathematical agent could keep a computationally supported conjecture separate from a formally verified theorem.

Codification also has limits. Existing criteria reflect disciplinary histories and distributions of power. They can exclude forms of

knowledge that do not fit dominant infrastructures, privilege questions that are easy to measure, and undervalue rare effects, local contexts, long time horizons, or Indigenous and experiential knowledge. An AI that follows today's rules perfectly may freeze today's exclusions. Scientific legitimacy cannot be reduced to compliance because science must retain the ability to revise its own criteria.

That reflexive capacity is among the hardest things to automate. An agent can learn current standards and generate alternatives, but changing what a community accepts as evidence involves values, risk, and authority. When a new instrument or method is accepted, researchers accept a new way of seeing. The telescope, microscope, controlled trial, statistical model, and computer simulation did not merely add observations. They changed the legitimate objects of knowledge. AI will do the same by making accessible patterns, synthetic objects, and representational spaces that people would not have constructed unaided.

The decisive test will not be whether those objects feel human. It will be whether they can enter a public economy of doubt. Can they be tested with different instruments? Can they generate risky predictions? Can they guide interventions that succeed outside the original laboratory? Can they be contested without permission from the model owner? When the answer is yes, science may expand its regimes of evidence. When the answer is no, we have technical authority rather than public knowledge.

No single AI evaluator can serve all these worlds. A system trained to detect p-hacking may misunderstand a proof. A benchmark critic may ignore clinical validity. A formal verifier may certify a program that implements a false empirical assumption. Serious automation requires domain-specific validation profiles connected to concrete ontologies, standards, and risks.

Moreover, the same result can justify different actions at different levels of maturity. An exploratory association may be enough to choose the next experiment but not to prescribe a treatment. A simulation may guide research funding but not determine public policy. Future scientific records will need intended-use statements: which conclusions the evidence supports and which decisions it must not yet authorize. Validity is not only a property of a sentence. It is a relation among a claim, the process that produced it, and the consequence for which it is invoked.

The Modern Knowledge Factory

3.1 From vocation to production system

The solitary genius remains science's most persistent portrait. It is a useful portrait because it makes discovery legible: one mind notices what others missed, persists through doubt, and changes the world. History contains people of extraordinary originality, but modern science is not mainly a collection of solitary minds. It is an infrastructure for coordinating partial knowledge. A result in particle physics can bear thousands of names. A biomedical discovery can depend on biobanks, clinical networks, technicians, software maintainers, statisticians, ethics committees, suppliers, regulators, and patients who consented to make their bodies part of an experiment. Even a theoretical paper written by one person rests on libraries, seminars, notation, public education, and generations of arguments.

During the twentieth century, science became a central organ of the state and the economy. War, medicine, agriculture, energy, industrial competition, and national prestige justified investments on a scale the gentleman experimentalists of the seventeenth century could scarcely have imagined. The Manhattan Project, national laboratories, space programmes, particle accelerators, weather satellites, and genome projects established what came to be called Big Science. Vannevar Bush's 1945 report, *Science, the Endless Frontier*, articulated an influential political compact: publicly supported basic research would replenish the reservoir from which prosperity, health, and security could draw.

This compact enlarged knowledge and changed the culture that produced it. The university became at once a school, a laboratory, an employer, a grant-seeking organization, and a node in an international competition for prestige. The number of researchers grew. So did the need to compare them from a distance. Publications, citations, journal impact factors, grant income, patents, rankings, and prizes became administrative proxies for a complex activity that could not be directly observed by every hiring or funding committee.

Metrics did not enter because administrators were foolish. They entered because large systems require compressed signals. Yet every compression creates a blind spot. Goodhart's law names the familiar danger: when a measure becomes a target, it ceases to be a good measure. Publication count can approximate productivity until careers depend on maximizing it; then one study becomes several papers, negative results vanish, and preliminary findings are dressed as breakthroughs. Citation counts can approximate influence until citation becomes currency; then strategic self-citation, fashionable topics, and dense mutual recognition networks flourish. Benchmark scores can approximate technical progress until laboratories train against the benchmark rather than the world.

These pressures do not imply that researchers are generally dishonest. They create ecologies in which reasonable individual choices can produce poor collective outcomes. A doctoral student who needs papers avoids a slow replication. A laboratory competing for renewal presents uncertainty in optimistic language. A journal seeking attention prefers a surprising claim. A university seeking rank rewards visible volume. The participants can remain sincere while the system produces a market in signals. Scientific culture is not outside society; it is one of society's institutions, with wages, hierarchies, deadlines, reputations, and scarcity.

Artificial intelligence enters this factory as both worker and manager. It can retrieve papers, summarize methods, clean data, generate code, optimize models, draw figures, draft prose, and monitor experiments. In a linear workflow, each saving may appear modest. In a chained workflow, savings compound: the output of literature search becomes the input to hypothesis generation, which becomes the input to code, which becomes the input to analysis and writing. A process that took months can, in a narrow and well-specified setting, be compressed into days or hours.

Errors compound in exactly the same way. A fabricated citation can shape an implausible hypothesis. The hypothesis can yield a technically correct but irrelevant experiment. A hidden data leak can produce an impressive figure. Fluent prose can explain the figure as though the result had been intended from the start. An automated reviewer can praise the clarity of the manuscript without discovering that the entire chain began from a false premise. Speed turns a local mistake into an assembly line.

For this reason, research productivity cannot be measured by the number of outputs an AI system produces. The proper object is the cost of validated knowledge. If an agent creates ten times as many manuscripts and doubles the burden of checking them, the scientific system may become less productive even though every dashboard rises. If it generates thousands of hypotheses but preserves neither their sources nor their rejected alternatives, it creates epistemic entropy: more claims, more apparent novelty, and less usable orientation. Valuable automation reduces the time to a robust claim and the cost of detecting error. It does not merely increase throughput.

A major empirical result published in 2026 makes this tension concrete. Hao and colleagues analysed 41.3 million papers across the natural sciences and found that researchers associated with AI-augmented work enjoyed striking professional advantages: higher publication and citation rates and earlier project leadership. At the collective level, however, AI adoption was associated with a contraction in the range of topics studied and reduced engagement among researchers. The tools appeared to pull attention toward areas already rich in data and amenable to computation (Hao et al., 2026). The finding does not prove that AI inevitably narrows science; observational studies of scientific careers cannot remove every confound. It does show that individual acceleration and collective exploration are different objectives. A system can help every user optimize locally while making the intellectual landscape more homogeneous.

This is an old social pattern in a new technical form. The Matthew effect, described by Robert Merton, captures how recognition accumulates around those already recognized. AI can add a computational flywheel. Laboratories with access to superior models, proprietary corpora, large compute budgets, robotic facilities, and specialist engineers produce more results. Those results attract more funding and data. The resulting systems improve faster, widening the velocity gap between institutions. Without shared infrastructure, automation may turn unequal resources into self-reinforcing differences in what can be known.

There is also an economy of attention. In a world of relatively few articles, a field can collectively read a meaningful fraction of its output. In a world of millions of AI-assisted manuscripts, visibility must be mediated by ranking systems. Those systems will influence which

questions are read, cited, funded, and replicated. Control over filtration can become as important as control over production. A company that does not own the laboratory but owns the scientific search, recommendation, and evaluation layer can quietly shape the frontier.

The role of the researcher will therefore change before it disappears. Some prestige will move away from manual execution toward problem formulation, constraint design, interpretation, audit, and the construction of reliable evidence pipelines. But institutions may continue rewarding visible novelty while leaving verification invisible. An AI-accelerated factory can produce abundance of results and scarcity of trust. Avoiding that outcome requires an economy that pays for the maintenance of truth: replication, curation, data stewardship, software preservation, calibration, red-teaming, and correction.

Big Science already taught researchers to depend on systems no individual fully understands. No single author grasps every subsystem of a collider, a climate model, or a genomic consortium. Legitimacy is distributed across organizational design, division of labour, standards, configuration management, and audit. AI does not create distributed cognition; it enters a world already built from it. The novelty is that some of the participating components can now generate plans, make choices, and write accounts of their own work.

That makes organizational design part of methodology. Who is permitted to launch an experiment? Which model may alter a protocol? What evidence causes a run to stop? Who owns the logs? Which failures are retained? How can an external group reproduce a result when the original system depended on a proprietary service that has since changed? These questions sound administrative, but they determine what the claim means. In automated science, governance is not a wrapper around method. It is one of the instruments through which method exists.

3.2 The reproducibility crisis as a warning

In 2015, the Open Science Collaboration reported an attempt to replicate one hundred studies from psychology. The effects observed in the replications were, on average, about half the size of the original effects. Whereas 97 percent of the original studies reported statistically significant results, 36 percent of the replications did; assessors judged that roughly 39 percent of the

effects had reproduced (Open Science Collaboration, 2015). The numbers became an emblem of a replication crisis, although the underlying problems were neither sudden nor confined to psychology.

A decade earlier, John Ioannidis had argued that in many research settings the combination of small samples, low statistical power, flexible analysis, selective publication, and a low prior probability of tested hypotheses made false or exaggerated findings likely to enter the literature (Ioannidis, 2005). Simmons, Nelson, and Simonsohn later showed how seemingly ordinary choices about sample size, outcome variables, covariates, and stopping rules could inflate false-positive rates when the analytical path was not constrained or fully disclosed (Simmons, Nelson, and Simonsohn, 2011). Similar concerns appeared in preclinical biomedicine, economics, cancer biology, and machine learning, although the mechanisms and standards differed by field.

The word crisis can suggest that science suddenly stopped working. A more accurate description is that old tensions became measurable. Statistical significance had been turned into a publication gate. Journals preferred positive and surprising results. Small studies produced unstable estimates. Data and code were often unavailable. Replication was slow, expensive, and professionally unrewarding. Researchers faced a garden of forking paths: many reasonable analyses could be chosen after seeing the data, and only the successful path might appear in the final paper.

The most constructive response was institutional rather than moralistic. Preregistration asked researchers to declare hypotheses and analysis plans before observing the relevant outcome. Registered Reports moved review before data collection, allowing publication to depend more on the importance of the question and quality of the design than on a spectacular result. Reporting standards such as CONSORT made missing information visible in clinical trials. The FAIR principles called for data to be findable, accessible, interoperable, and reusable (Wilkinson et al., 2016). Reform proposals emphasized statistical power, openness, collaboration, uncertainty, and changes to incentives (Munafò et al., 2017).

None of these practices guarantees truth. Preregistration cannot rescue a weak question. Open data can expose an analysis without making it correct. A Registered Report can yield a rigorous answer to an unimportant question. FAIR data can still embody biased sampling.

Reproducibility is not a binary seal attached to a paper. It is an ecology of conditions that makes error easier to discover and correction less costly.

AI arrives after science has paid dearly for this lesson. It would be perverse to automate the old workflow at high speed and add transparency only when failures become public. A research agent should be designed natively for versioning, provenance, preregistration, and tests. Analytical choices should be timestamped before the agent sees the score they could improve. Hyperparameter search should be separated from final evaluation. Failed experiments should remain in the record rather than disappear from the agent's memory. Every claim should be linked to data, code, environment, and the transformations that support it.

Automation can make reproducibility easier. Agents can generate containers, lock dependencies, record random seeds, rerun analyses, test data schemas, detect discrepancies between text and code, and produce machine-readable protocols. A laboratory robot can log temperatures, timings, reagent lots, instrument states, and deviations with a consistency no human notebook can match. An automated system can preserve the search path rather than merely the winning result.

Automation can also make reproducibility harder. Foundation models change. Hosted services are proprietary. Outputs are stochastic. Retrieval indices are updated. Tool-routing policies, hidden prompts, safety filters, and memory stores can alter a trajectory. Reproducing an agentic result is not simply rerunning a script. It may require preserving the model version, context, retrieved documents, tool schemas, permissions, random state, hardware, external services, and exact order of actions. The method becomes a distributed digital object.

A human laboratory notebook could be incomplete, but at least it had recognizable boundaries. An agent can execute hundreds of calls, spawn subtasks, revise files, and branch into alternatives. Without observability, the result has no recoverable history. Without history, it cannot be audited. Without audit, reproducibility becomes a marketing word.

The strongest architecture separates exploration from confirmation. During exploration, an agent may search freely, generate many hypotheses, and tune systems against provisional data. Before confirmation, the intended claim, primary outcome, exclusion rules,

analysis, stopping criteria, and permitted adaptations are frozen. The confirmatory run uses held-out data or a fresh experiment that the searching agent could not inspect. Deviations are allowed when reality requires them, but they are recorded and interpreted as deviations. This is preregistration translated from a document into an access-control system.

The distinction matters because an AI can comply with the letter of openness while exploiting the space before the log begins. A system might try thousands of specifications, retain the most promising, and then preregister it as though it had been predicted. It might use benchmark feedback indirectly through public leaderboards. It might delete failed branches from the visible workspace. Full provenance must therefore include the relevant search history and data-access history, not only the polished route to the result.

Reproduction also has degrees. Running the same code on the same data checks computational repeatability. Reimplementing the method checks whether the description is sufficient and the result is not an implementation accident. Collecting new data tests empirical robustness. Reproducing across laboratories, instruments, populations, suppliers, or environments tests transportability. Future records should state which level has been achieved. The word replicated cannot continue to cover every act from pressing the same button twice to independently recreating the phenomenon.

Benchmarks of autonomous research show why this discipline is urgent. PaperBench asks agents to reproduce twenty machine-learning papers through 8,316 rubric-scored subtasks. In the original evaluation, the best tested agent reached an average score of 21 percent, well below the recruited expert baseline (Starace et al., 2025). ResearchClawBench, introduced in 2026 across forty tasks and ten scientific domains, similarly found that leading systems remained around the low twenties on end-to-end rediscovery, with errors concentrated in protocol mismatch, evidence mismatch, and omission of the scientific core (Xu et al., 2026). AutoResearchBench found single-digit performance on demanding deep and wide literature-discovery tasks even when general browsing benchmarks had become much easier for frontier models (Xiong et al., 2026). These benchmarks are imperfect proxies for science, but their failure analyses are revealing: present agents are often better at producing

plausible activity than at preserving the exact relation between question, method, and evidence.

The reproducibility crisis offers a lesson in institutional humility. Science did not lose all legitimacy when researchers discovered that many published results were fragile. Its legitimacy was strengthened where communities organized replications, exposed failure, and altered practice. A culture that claims infallibility is brittle. A culture that instruments correction can survive error. Artificial researchers must be introduced into the second culture, not the first.

3.3 Science and scientism in the age of infinite text

Science is a family of practices for producing and correcting empirical and formal knowledge. Scientism is the unjustified extension of scientific authority beyond the domains in which those practices have earned it. Mikael Stenmark distinguished several forms: the claim that science is the only source of genuine knowledge; that only scientifically recognized entities are real; that values can be derived from science alone; or that scientific progress will ultimately redeem every human problem (Stenmark, 1997). Susan Haack identified characteristic symptoms: using the word scientific as an honorific, imitating technical surfaces without corresponding discipline, and demanding a single boundary that cleanly separates science from every other form of understanding (Haack, 2012).

AI can strengthen scientism because it makes the signs of scientificity almost free. A model can generate equations, polished graphs, confidence intervals, formal definitions, cautious prose, and extensive bibliographies. It can convert a political preference into an optimization problem, attach decimals to a poorly defined judgement, or produce a literature review that appears exhaustive. Pseudoscience once required laborious imitation. It can now be manufactured at industrial scale.

The danger does not come only from obvious charlatans. Legitimate institutions can misuse the same authority. A public administration can present a value choice as the inevitable output of a model. A company can turn its commercial objective into a recommendation described as data-driven. A scientific community can treat the most retrievable literature as the entire state of knowledge. When an agent synthesizes

millions of documents, the size of the corpus creates an aura of totality. Yet the corpus still omits unpublished failures, inaccessible papers, private data, marginalized languages, local observation, tacit knowledge, and questions nobody had resources to ask.

Messeri and Crockett call attention to illusions of understanding in AI-assisted science: researchers may believe they see more broadly, work more objectively, or understand more deeply when tools actually homogenize viewpoints and conceal gaps (Messeri and Crockett, 2024). Fluency intensifies the illusion. Difficulty used to provide a useful signal that a concept had not been mastered. A model can remove the felt difficulty while leaving the conceptual gap untouched.

The liturgy of algorithmic scientism is already easy to imagine. The model analysed all the data. The agent checked the literature. The objective was optimized. The simulation proved the policy. Each sentence can be valid under restricted conditions and abusive outside them. Analysing data does not prove the variables were well chosen. Checking literature does not reveal what was never published. Optimization does not justify the objective. Simulation does not establish that the model contains every politically relevant feature of the world.

Citation is especially vulnerable to symbolic inflation. OpenScholar's evaluation demonstrated that retrieval and citation-aware training can greatly improve scientific synthesis, but it also quantified the weakness of ungrounded generation. In its tests, general models asked to cite recent literature frequently produced plausible titles that did not exist, and even real references often failed to support the adjacent claim (Asai et al., 2026). The moral is not that generated text is useless. It is that the visual form of scholarship has become separable from scholarship's evidential function.

Scientism will become most dangerous when combined with opacity and institutional convenience. A decision-maker can say that the system is too complex to explain, too accurate to challenge, and too efficient to replace. Responsibility dissolves into the pipeline: the scientist points to the model, the model owner points to the data, the administrator points to the performance metric, and the harmed person is told that no individual chose the outcome. This would be a return of priestly authority in technical clothing.

A critique of scientism is not a rejection of science. It protects science from becoming ideology. Facts constrain action, but they do not

uniquely determine aims. Climate models can estimate consequences; they do not decide the just distribution of costs. Epidemiological models can compare interventions; they do not define the acceptable balance between liberty, vulnerability, and collective risk. Genetic evidence can characterize probabilities; it does not decide which lives merit support. Values do not disappear when numbers arrive. They become easier to hide inside objective functions.

The semiotics of AI-generated science makes this hiding especially effective. Decimal places, uncertainty bands, dense citations, and technical diagrams are signs associated with disciplined inquiry. Their presence can substitute psychologically for the process they normally signify. A confidence interval may be computed correctly around an invalid measurement. A visualization may faithfully display a biased sample. A formal proof may certify a program whose empirical premises are false. The sign is not fraudulent in itself; the fraud lies in allowing it to carry more authority than the chain beneath it can support.

Society will therefore need a new scientific literacy. Citizens need not reproduce every derivation, but they should be able to ask what data were included, what objective was optimized, which alternatives were excluded, what uncertainty remains, what decision the evidence does and does not support, and who is answerable. Researchers must learn to distinguish a tool that enlarges observation from one that narrows imagination. Institutions must disclose not only performance but scope, dependencies, conflicts, and known failure modes.

The question how do we distinguish the charlatan from the scientist will no longer be answered by prose style, credentials, or the presence of citations. Both can be generated or rented. The distinction will increasingly depend on conduct: whether claims are linked to inspectable evidence; whether uncertainty is preserved; whether failures are reported; whether independent parties can test the result; whether criticism changes the record; whether incentives are disclosed; and whether the speaker accepts conditions under which the claim would be withdrawn.

Charlatanism is not simply being wrong. Scientists are often wrong. It is the systematic insulation of a claim from correction while borrowing the symbols of corrigibility. An AI system can become a perfect machine for that insulation if it creates endless explanations after every failure. Conversely, the same system can strengthen science if it records predictions in advance, searches for disconfirming evidence, exposes alternative analyses, and makes retraction technically and socially possible.

The scientific culture of the future should therefore become more modest precisely as its tools become more powerful. Capability increases the obligation of contestability. A system that can read more than any person must expose what it could not access. A system that can search millions of hypotheses must reveal the search and protect a genuine test set. A system that can act in the world must accept stronger permissioning and independent oversight. The opposite principle - that performance earns exemption from scrutiny - would convert research automation into automated authority.

When the Tool Began to Propose

4.1 From index to hypothesis

Research automation did not begin with conversational models. It has a long genealogy made from bibliographic indexes, statistics, expert systems, optimization, scientific software, and laboratory robotics. Each stage moved part of cognition into an artifact. The table externalized comparison. The graph made relations visible. The calculator accelerated operations. The database separated personal memory from access to a field. The search engine transformed discovery into ranking. Long before a machine could compose an explanation, instruments had already begun deciding what could be seen.

One of the clearest ancestors of today's research agents appeared in the 1980s. Don Swanson noticed that two literatures could jointly contain an hypothesis that neither had stated. Papers about fish oil discussed effects on blood viscosity, while papers about Raynaud's syndrome described circulatory problems. The bridge suggested a possible intervention. Swanson called such connections undiscovered public knowledge. The method later became known through the ABC pattern: when one literature relates A to B and another relates B to C, an A-C relation may deserve investigation (Swanson, 1986).

The pattern seems elementary, but it changes the locus of discovery. The new hypothesis need not arise from a new observation. It can arise from a new path through observations that were already public but socially separated. Disciplinary boundaries, vocabulary, and limited human attention had prevented the connection. Literature-based discovery made ignorance partly a graph problem.

Contemporary knowledge graphs elaborate this intuition. They represent papers, authors, compounds, genes, methods, measurements, datasets, claims, and relations as nodes and edges. An algorithm can search for missing links, contradictory paths, clusters, or unusually distant analogies. Unlike a language model's continuous internal representation, a knowledge graph can expose at least part of its structure: the entities it believes exist, the relations it asserts, and the

sources from which they were extracted. This explicitness is valuable for provenance, but it is not neutral. Ontologies decide what kinds of things are allowed to exist in the representation. Entity extraction can merge different concepts or separate synonyms. A graph can make a false relation look exact.

The next decisive step was closing part of the loop between hypothesis and experiment. In 2004, a system described as a robot scientist automated cycles of experimentation, and the later system Adam generated hypotheses about yeast gene functions, designed and executed experiments, interpreted results, and represented its work in a formal language (King et al., 2004; King et al., 2009). Adam was narrow. Its domain, instruments, and hypothesis language were carefully structured. That narrowness was not a defect in the demonstration; it was the condition of success. A portion of scientific method became executable because the world had first been reduced to a tractable set of entities, operations, and measurements.

AutoML automated another region of scientific labour. Data preprocessing, algorithm selection, hyperparameter tuning, neural architecture search, and ensembling became search problems. Systems such as auto-sklearn, TPOT, AutoGluon, FLAML, H2O AutoML, and commercial cloud services can build strong predictive baselines with less manual intervention. In a well-defined task, they force comparison against a serious reference rather than a hand-tuned favourite. Yet the loss function cannot tell the system whether the target was measured well, whether the dataset reflects the population, whether deployment changes behaviour, whether causality is required, or whether the question matters.

This distinction separates automation of optimization from automation of inquiry. AutoML can answer, under specified constraints, which pipeline scores best on a given dataset. Science must also ask why that dataset exists, what phenomenon the labels operationalize, which alternative explanations survive, and what action the score licenses. A machine can be excellent at moving inside a problem definition and powerless to judge the definition itself.

Alongside AutoML grew a broader family often called scientific machine learning. Its purpose is not primarily to manage a research workflow but to learn usable models of natural or engineered systems. Surrogate models, also called emulators, approximate an expensive simulation or experiment. Physics-informed networks incorporate

equations, symmetries, conservation laws, or boundary conditions into learning. Neural operators learn mappings between whole functions rather than between fixed vectors. Differentiable simulators let a design be optimized by propagating gradients through a model of the world. Foundation models for proteins, molecules, materials, climate, and microscopy transfer patterns learned at scale into specialised scientific tasks (Karniadakis et al., 2021).

These systems can change the economics of inquiry before they become agents. AlphaFold made a difficult structural prediction available in minutes or hours rather than months of experimental work in favourable cases (Jumper et al., 2021). GraphCast learned a fast model of global weather evolution from reanalysis data and demonstrated that a learned surrogate can compete with an operational numerical forecast on many targets (Lam et al., 2023). GNoME used graph networks and an active-learning cycle to search an immense crystal space, providing candidates later connected to laboratory synthesis (Merchant et al., 2023). In each example, the machine does not conduct the entire scientific process. It creates a new instrument of inference that changes where experiments, human attention, and verification are spent.

The opportunity is enormous. A trusted emulator can support uncertainty analysis that would be too expensive with the original solver, screen millions of candidates before physical testing, or let a robot choose experiments in real time. Inverse design reverses the usual direction of science: instead of asking what properties a given object has, the system searches for an object that would have desired properties. When such models are linked to agents and laboratories, prediction becomes one layer of an automated discovery engine.

The danger is equally structural. A surrogate can be exquisitely accurate inside its training regime and fail smoothly outside it. A physics-informed model inherits the equations and boundary conditions supplied by its designers; it cannot discover a missing process if that process has been excluded from both data and constraints. A foundation model can compress historical regularities while reproducing the blind spots of the corpus. Scientific machine learning therefore needs domain-of-validity tests, uncertainty calibration, stress tests, conservation checks, and contact with fresh measurements. It accelerates the production of hypotheses; it does not abolish the obligation to confront the world.

This distinction clarifies two phrases that are often confused. AI for science uses machine learning to model, predict, simulate, classify, or design within a scientific domain. An AI scientist coordinates parts of inquiry: it searches evidence, proposes questions, designs tests, executes tools, evaluates outcomes, and decides what to do next. The first can be revolutionary without being autonomous. The second is impossible to trust unless the models it orchestrates expose their limits.

Systematic review automation offers a parallel lesson. The work of evidence synthesis includes formulating a question, designing searches, deduplicating records, screening titles and abstracts, reading full texts, assessing bias, extracting data, comparing populations and interventions, performing meta-analysis, interpreting heterogeneity, and writing conclusions. Tools such as RobotReviewer and ASReview have shown that some stages can be accelerated. ASReview uses active learning: a human labels records, the model learns which papers are likely to be relevant, and the next records are prioritized. The point is not to replace the reviewer with a classifier but to spend scarce human attention where it has the highest expected value (van de Schoot et al., 2021).

Recent surveys of automated evidence synthesis show that this remains mostly stage automation rather than full methodological substitution. Screening and extraction are easier to automate than assessing whether studies are meaningfully comparable, whether an outcome is clinically relevant, or whether heterogeneity invalidates a pooled estimate. Large language models widen the interface, but they do not remove the difference between finding text and evaluating evidence. A system may extract every reported number correctly and still combine studies that should never have been treated as one experiment.

The arrival of large language models connected these traditions through a general semantic interface. A natural-language objective can be translated into queries, code, tool calls, protocols, or requests to a robot. The first enthusiasm often imagined the model as a universal memory. Scientific use quickly exposed the weakness of that picture. Parametric knowledge becomes stale, rare details are unreliable, citations can be fabricated, and confidence is poorly calibrated. Retrieval-augmented generation offered a more credible pattern: the model searches an external corpus, brings relevant passages into

context, and generates an answer linked to evidence (Lewis et al., 2020).

RAG changes the aspiration from a model that knows to a system that knows how to look and how to show what it used. In research, however, retrieval is not a single operation. The agent must formulate queries, select databases, traverse citations, distinguish versions, expand terminology, recognize negative evidence, compare primary and secondary sources, and decide when the search is saturated. It must preserve the path so that a later reader can see why some literature entered the synthesis and some did not.

A large language model without tools is primarily a generator conditioned on compressed linguistic regularities. With retrieval, code execution, theorem provers, simulators, databases, and instruments, it becomes an orchestrator. The model does not need to contain the whole of science. It needs to select specialized operations, interpret their outputs, preserve a goal across time, and know when uncertainty requires escalation.

This is the origin of the scientific agent. Unlike a chatbot, an agent has a persistent task, state, memory, tools, a policy for choosing actions, and feedback that can modify its plan. It may decompose a question, retrieve literature, formulate hypotheses, write code, inspect errors, run experiments, compare results, and draft a report. Multi-agent systems divide these functions into roles: generator, critic, planner, statistician, literature scout, experimentalist, safety officer, or reviewer. The architecture attempts to internalize something like the division of labour of a research group.

The analogy can mislead. Several roles produced by the same model are not several independent minds. They can share training data, blind spots, stylistic preferences, and hidden assumptions. A debate among prompts may converge elegantly on a false premise. Independence requires diversity of evidence, implementation, model lineage, incentives, and physical realization. A critic that receives the author's entire chain and is rewarded for agreement is not an external reviewer merely because its system prompt says sceptic.

Across this history, the verbs assigned to machines have changed. First the machine calculated. Then it stored, searched, classified, and optimized. Now it proposes, plans, executes, interprets, and publishes. Each verb transfers a different kind of power. A bad calculation produces a bad number. A bad proposal can orient an entire

programme. A bad action can consume resources, damage equipment, expose confidential data, or create physical risk. As autonomy increases, evaluation must move from the correctness of isolated answers to the safety and epistemic integrity of trajectories.

The most general pattern behind recent successes is not a particular model. It is the coupling of a generator to an evaluator and a memory. FunSearch generated programs and selected them through executable scoring, producing new results in combinatorial mathematics (Romera-Paredes et al., 2024). AlphaGeometry combined a language model with a symbolic deduction engine to solve difficult geometry problems (Trinh et al., 2024). Systems for symbolic regression, equation discovery, theorem proving, and empirical software use different representations, but their strongest results arise when candidate production is cheap and checking is severe.

This pattern reveals where automation will advance fastest. A domain becomes tractable when it has a rich search space, a generator capable of exploring it, an evaluator that is cheaper and more reliable than generation, and a memory that prevents repeated failure. Mathematics supplies proof checkers. Programming supplies tests and execution. Games supply rules and scores. Materials and chemistry can supply instrument readings. The social sciences and medicine can also supply measurements, but the relation between metric and human meaning is often more contested. The frontier is therefore not distributed evenly across disciplines.

4.2 The research-automation stack

The most credible systems of the mid-2020s do not treat the language model as an oracle. They use it as a semantic control layer inside a stack. At the bottom are the world and its records: literature, databases, instruments, samples, simulations, and people. Above them are tools that retrieve, transform, calculate, and act. The language model coordinates those tools. Around the entire process sit provenance, evaluation, safety, and governance. State of the art is not a single intelligence. It is a composition.

The research-automation stack

Capability grows upward; provenance and governance span the system.

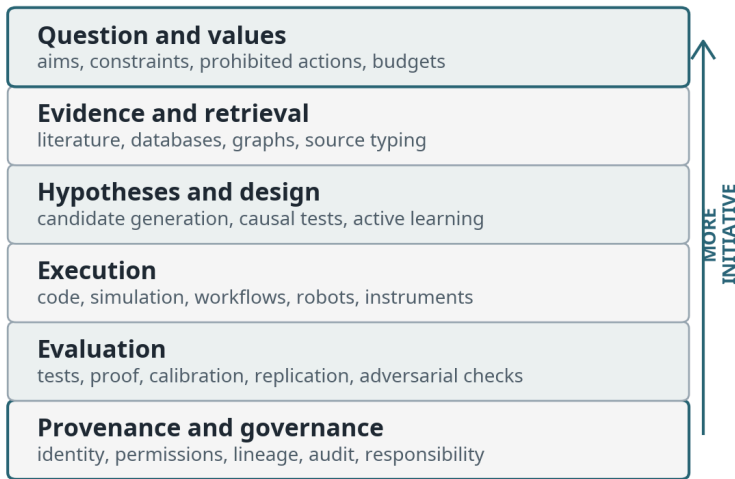


Figure 2. The research-automation stack: from a question to evidence, execution, and independent verification.

Every research campaign begins with problem formalization. An instruction such as find a more stable compound is not operational until stability, temperature, duration, cost, toxicity, manufacturability, and acceptable trade-offs are defined. The same ambiguity that a human team might resolve through conversation becomes executable ambiguity in an agent. Safe systems therefore begin with objectives, constraints, prohibited actions, budgets, stopping conditions, and an explicit statement of who may alter them.

The next layer is epistemic intake. This includes literature retrieval, database queries, citation-graph traversal, patent search, data discovery, and, increasingly, multimodal search across tables, figures, spectra, images, and code. Open infrastructures such as Crossref, OpenAlex, PubMed, Europe PMC, Semantic Scholar, and domain repositories provide parts of the public memory. Commercial tools such as Elicit, Consensus, Scite, Litmaps, ResearchRabbit, and publisher platforms offer useful interfaces over parts of the same landscape. Their capabilities change quickly, and a feature list becomes obsolete. The stable issue is architectural: what corpus can the system see, what

ranking logic governs visibility, what metadata is retained, and whether the user can export the evidence trail.

Retrieval must be followed by evidence typing. A search result is not yet evidence. The system needs to distinguish a randomized trial from an observational study, a preprint from a corrected version of record, a data paper from an inference, a methodological benchmark from a deployment study, a retraction from a citation that failed to notice the retraction, and a review's interpretation from the primary experiment. In mature systems, claims do not simply point to documents. They point to passages, tables, figures, data, and methodological properties of those sources.

OpenScholar demonstrates the value of specializing this layer. Its corpus contains tens of millions of open-access papers, with trained retrieval, reranking, iterative feedback, and citation verification. On ScholarQABench, a smaller specialized model could outperform much larger general systems on difficult multi-paper synthesis, while markedly improving citation accuracy (Asai et al., 2026). The lesson is not that literature synthesis is solved. It is that research quality depends as much on the evidence system as on the generative model.

The open-access corpus also reveals a structural limit. Reproducibility improves when the corpus can be inspected and redistributed, but the resulting system does not see paywalled literature, confidential industrial studies, some books, local reports, or knowledge never written in a machine-readable form. Retrieval can create a precise map of an incomplete territory. An honest agent must describe the boundary of the map.

Synthesis comes next. The system compares methods, populations, measurements, assumptions, results, and limitations. It can cluster positions, identify contradictions, and create a provisional state of the art. Yet synthesis is an inference, not a collage. A paragraph can be supported sentence by sentence and still misrepresent the field through selection. Coverage, not merely citation correctness, matters. So do temporal order, publication bias, conceptual drift, and the weight assigned to different study designs.

Knowledge graphs can stabilize some of this structure. They can represent that one paper uses a dataset, that a dataset measures a construct, that a method depends on an assumption, or that a result contradicts another under a specific condition. Graphs are especially valuable when an agent must reason across distant literatures or

preserve provenance over long projects. Their weakness is brittleness: scientific language is ambiguous, relations are context-dependent, and ontology construction can turn a disputed concept into a hard edge. The best systems combine graph structure with access to the underlying text and data rather than treating either as sufficient.

Hypothesis generation then transforms synthesis into search. An agent can propose dozens or thousands of candidate explanations, interventions, algorithms, or experiments. Multi-agent systems may debate them, rank them through tournaments, mutate them, and seek analogies from other fields. Google's Co-Scientist uses specialized agents for generation, reflection, ranking, evolution, proximity, and meta-review under a supervisory process. In biomedical evaluations, the system generated proposals that experts judged promising, and selected hypotheses were tested experimentally (Gottweis et al., 2026). Robin integrated literature and data-analysis agents to propose and update biological hypotheses in a semi-autonomous cycle (Ghareeb et al., 2026).

The striking capability is not that the systems have human-like inspiration. It is that they can maintain a high-volume evolutionary process over ideas. Candidate generation, criticism, comparison, and revision occur repeatedly. This can surface combinations that a small team would not consider. It can also produce an ocean of plausible variations. When ideas become cheap, novelty checking, prioritization, and experimental discrimination become the scarce resources.

Novelty itself must be treated cautiously. A phrase can be new while the underlying relation is old. A proposal may be absent from published papers because it failed repeatedly, was patented, remained in laboratory notebooks, or was rejected as unsafe. Search against literature and patents gives only a lower bound. The strongest agents should represent novelty as an uncertain claim, state the corpora examined, and separate semantic difference from substantive difference.

Experiment design translates a hypothesis into an opportunity to be wrong. Here agents can choose controls, sample sizes, variables, measurement schedules, and discriminating conditions. Statistical design tools already automate power analysis and adaptive allocation. Bayesian optimization and active learning select the next experiment expected to reduce uncertainty or improve an objective. Causal

discovery systems propose graphs or interventions. Digital twins simulate possibilities before physical execution.

This layer exposes a deep difference between prediction and science. A prediction system seeks low error. An experiment seeks information capable of distinguishing explanations. The most useful next observation may not maximize immediate performance; it may visit a region where rival models disagree. An autonomous scientist needs objectives such as information gain, robustness, mechanism discrimination, and value of information, not only task score.

Execution turns plans into artifacts. In computational science, the agent writes code, constructs environments, downloads data, runs jobs, interprets failures, and modifies the implementation. In experimental science, it translates protocols into commands for robots and instruments. Sandboxing, least privilege, resource limits, network controls, and chemical or biological safety constraints are not merely cybersecurity features. They preserve the meaning of the experiment. An uncontrolled agent can alter data, leak test information, call unapproved services, or achieve a score through an unintended side effect.

Long-horizon systems increasingly use durable workspaces rather than relying on conversational memory alone. Plans, code, data dictionaries, logs, hypotheses, and decisions become files or structured objects that specialized agents can inspect. This pattern can be summarized as thin control over thick state: the orchestrator carries a compact view of the project while the full evidence resides in persistent artifacts. It is closer to a real laboratory than a long chat transcript because responsibility can be attached to objects and versions.

Evaluation closes the loop. Where a reliable scalar exists, tree search, evolutionary algorithms, Bayesian optimization, or reinforcement learning can search aggressively. ERA combines language-model code generation with tree search over scorable empirical tasks. It reported new methods for single-cell analysis, forecasting, geospatial reasoning, neural prediction, and computational mathematics that matched or surpassed strong human-developed references (Aygün et al., 2026). The system is most powerful where an evaluator can cheaply distinguish improvement from decline.

Not every scientific value has a faithful scalar. Novelty, explanatory depth, clinical relevance, social benefit, and ethical acceptability are multidimensional and contested. Collapsing them into one number

creates a target for Goodhart's law. A robust evaluator preserves a vector of properties and the disagreements among judges. It can report predictive performance, calibration, computational cost, robustness, fairness, interpretability, and external validity separately. The decision to trade one for another remains visible.

Memory and provenance span all layers. Each hypothesis needs a lineage. Each source needs a version. Each transformation needs an identifier. Each run needs an environment, seed, hardware description, input hash, and output. Each human intervention needs a record. Each tool response needs enough context to be interpreted later. A brilliant agent without verifiable memory resembles a virtuoso who burns the notebook after every experiment.

Governance surrounds the stack. A system that can retrieve papers is an assistant. A system that can execute code is an operator. A system that can spend money, access sensitive data, control instruments, contact participants, submit manuscripts, or allocate laboratory time is an institutional actor. Permission must increase in stages, with stronger evidence required for higher-impact actions. The right to propose should not imply the right to execute, and the right to execute should not imply the right to publish or deploy.

A useful taxonomy therefore classifies research automation along several independent dimensions. One dimension is the stage automated: search, synthesis, hypothesis, design, execution, analysis, writing, review, or dissemination. Another is the degree of initiative: tool, copilot, delegated agent, supervised team, or autonomous programme. A third is the environment: text, code, simulation, data infrastructure, or physical laboratory. A fourth is the evaluator: human judgement, benchmark, statistical test, formal proof, instrument, market response, or social outcome. A fifth is the evidence boundary: what the system can access and what remains invisible. A sixth is the consequence of failure.

This multidimensional view is more honest than a label such as AI scientist. A system may be highly autonomous in code generation and entirely dependent on a human for problem selection. It may conduct physical experiments autonomously inside a narrow materials platform while having no ability to interpret social value. It may write a complete paper while being unable to validate a central claim. Autonomy is local before it is general.

Table 2. Representative research-automation systems and their present limits.

System family	Capability and limiting condition
Literature synthesis (OpenScholar, PaperQA2, ASReview)	Strong retrieval and screening; bounded by corpus coverage, ranking, and evidence appraisal.
Hypothesis teams (Co-Scientist, Robin)	Broad proposal and debate; independence and experimental validation remain limiting.
Empirical software (ERA, AutoML)	Powerful where the objective is machine-scorable; metric quality defines success.
End-to-end research (AI Scientist, Agent Laboratory)	Integrates the workflow; brittle on long horizons and independent confirmation.
Self-driving labs (Coscientist, A-Lab, ChemOS)	Closes the physical loop; safety, calibration, and material variability dominate.
Benchmarks (PaperBench, ResearchClawBench)	Make claims measurable; current systems remain far from reliable general autonomy.

The open-source versus commercial distinction cuts across the stack. Open systems permit inspection, adaptation, and independent hosting, but may lack polished interfaces, licensed corpora, compute, or sustained maintenance. Commercial systems can integrate search, models, proprietary databases, and support, but may obscure ranking, prompts, versions, data use, and evaluation. Scientific institutions should avoid making the long-term record dependent on a service that cannot export its provenance. Convenience at the interface can become fragility in the archive.

The most reliable pattern is not complete replacement but calibrated delegation. The agent does what is fast, repetitive, searchable, or executable; the researcher defines aims, inspects assumptions, adjudicates ambiguous evidence, and accepts responsibility. As systems improve, the boundary moves. But a moving boundary is not the disappearance of human judgement. It is a demand to place judgement where the system's evaluator is weakest and the consequences are largest.

4.3 Measured promise, visible limits

Popular discussion often asks whether an AI can do science. The empirical answer depends on the granularity of the task. On tightly scored problems, systems can exceed expert baselines. On open, long-horizon projects, they remain brittle. Both statements are true, and their coexistence is the most important fact about the current frontier.

The AI Scientist offers a vivid end-to-end demonstration. Its pipeline generates ideas, writes and modifies code, runs experiments, produces figures, drafts manuscripts, and applies an automated review process. A later version reduced dependence on hand-supplied experimental templates and used tree-structured search. In work published in *Nature* in 2026, an AI-generated manuscript passed the first round of review for a machine-learning workshop (Lu et al., 2026). The achievement shows that a substantial fraction of a familiar research genre can be composed into an automated pipeline.

It does not show that general autonomous science has arrived. The domain was machine learning, where experiments can be executed cheaply and evaluated numerically. The generated papers varied in quality. Human researchers still chose systems, budgets, submission decisions, and evaluation conditions. Workshop acceptance is evidence that a manuscript crossed a social threshold, not that every claim was profound or independently replicated. The correct conclusion is neither dismissal nor prophecy: the surface and part of the substance of a research cycle are now automatable under favourable conditions.

Agent Laboratory and related frameworks show a similar pattern through multi-agent division of labour. Distinct roles review literature, plan experiments, write code, analyse results, and draft papers (Schmidgall et al., 2025). AgentRxiv imagines an environment in which agents publish and iteratively improve one another's work. Agent4Science, reported in 2026, even created a social space in which artificial agents could share and debate research while humans observed (Ahart, 2026). These projects are valuable not because their communities already possess the institutional depth of science, but because they make a future governance problem concrete. Citation, debate, and publication can be simulated before epistemic independence has been achieved.

CellVoyager illustrates progress in a domain with complex biological data. The system autonomously analysed single-cell data and generated findings later examined by human scientists (Alber et al., 2026). BioMedAgent coordinated specialized tools across biomedical workflows and achieved strong performance on a task benchmark (Bu et al., 2026). Co-Scientist and Robin moved from literature toward hypothesis and experimental interpretation. These systems suggest that the near-term frontier is not a single universal agent but a general coordinator surrounded by domain tools and human experts.

Benchmarks that demand reconstruction rather than polished text provide a more severe picture. PaperBench requires agents to reproduce twenty ICML papers from scratch and evaluates them through thousands of author-informed rubric items. In the original report, the strongest tested configuration averaged 21 percent and did not beat the expert human baseline (Starace et al., 2025). MLE-bench asked agents to solve seventy-five Kaggle competitions; early systems achieved a bronze-level result in only a minority of tasks. ScienceAgentBench found that leading agents could independently complete only about a third of diverse data-driven research tasks in its initial evaluation. MLR-Bench reported a particularly troubling failure mode: coding agents frequently produced fabricated or invalid experimental claims even when their repositories appeared plausible (Chen et al., 2025).

The newer benchmarks of 2026 make the same weakness visible at greater scale. AutoResearchBench found that frontier systems achieved roughly nine percent on demanding literature-discovery tasks. ResearchClawBench placed agents in forty end-to-end scientific tasks across ten domains and found leading scores around the low twenties, with failures concentrated in experimental protocol mismatch, evidence mismatch, and missing the scientific core (Xiong et al., 2026; Xu et al., 2026). AutoMedBench found that agents were better at making a pipeline run than at validating whether it was reliable: verification and submission failures dominated its error analysis (Liu et al., 2026).

These measurements should be interpreted with care. Benchmarks sample a narrow and artificial slice of research. Human scientists also need documentation, collaboration, prior familiarity, and time. An agent may fail because a benchmark hides resources a real laboratory would possess. Conversely, benchmark tasks are cleaner than open science and may overestimate transfer. Scores can be gamed once

training systems see the rubrics. The purpose of the benchmarks is not to assign a single intelligence number. It is to locate failure modes and make claims of autonomy falsifiable.

The contrast between local excellence and end-to-end fragility follows a simple mathematical intuition. Research is conjunctive. A successful project may require adequate literature search, correct environment setup, valid data handling, appropriate design, bug-free code, faithful analysis, calibrated interpretation, and honest reporting. If each stage succeeds with a probability of nine-tenths, eight dependent stages yield a whole-project probability far below nine-tenths. Real failures are correlated, making the chain more fragile. A fluent report at the end can conceal the missing link.

Current systems also struggle with metacognition. They often cannot reliably recognize that they do not know, distinguish a shallow answer from a verified one, or calibrate confidence under distribution shift. Self-critique can improve outputs, but the critic may share the generator's error. More inference-time computation can deepen search, yet it can also produce more elaborate rationalization. Scientific autonomy requires not only competence but the ability to detect when competence has ended.

The most consequential opportunity lies in hybrid systems that connect complementary strengths. Machines can read broadly, remember exactly, search tirelessly, generate many candidates, execute repeatable operations, and monitor complex processes. Humans can choose questions in relation to lived needs, notice when a formalization has lost the phenomenon, interpret tacit context, negotiate values, and bear public responsibility. Neither list is permanent. What matters is designing the interface so that strength does not conceal weakness.

A mature deployment begins by asking what property is being delegated. Search can be evaluated through recall and source quality. Code can be evaluated through tests, held-out data, and inspection. A mathematical derivation can be checked formally. A laboratory action can be constrained by interlocks and monitored by independent sensors. A clinical interpretation may require domain expertise, patient context, and regulatory accountability. The phrase human in the loop is too vague. The human must have a defined decision, sufficient information, genuine authority, and time to intervene.

Automation will therefore advance through islands of high autonomy connected by bridges of human and institutional judgement. Literature triage, routine analysis, parameter search, instrument scheduling, and reproducibility checks may become almost invisible services. Question selection, high-risk experimentation, interpretation across value conflicts, and authorization of social consequences will remain more contested. Over time, some islands will merge. The bridges must be redesigned rather than removed by default.

The frontier is best described as a change in the unit of research. The unit is no longer only the human investigator or the software tool. It is the human-agent-tool-institution system. Its performance depends on architecture, incentives, evidence access, and governance. Asking whether the model alone is a scientist is like asking whether the air pump alone discovered the vacuum. The apparatus matters profoundly, but the scientific object emerges from an organized relation among apparatus, witnesses, procedures, and a community capable of correction.

The next step gives the agent a body. Once hypotheses can control pumps, pipettes, beamlines, microscopes, and reactors, the distance between sentence and consequence collapses. The research assistant becomes an experimental operator, and the ancient workshop returns in automated form.

The Laboratory That Does Not Sleep

5.1 When the hypothesis grows arms

A model that writes a wrong answer can mislead. A model connected to a robot can turn error into action. This passage from language to the physical world is the point at which research automation becomes cyber-physical. The agent is no longer only an interlocutor. It becomes a component of the laboratory, able to select reagents, schedule instruments, interpret sensors, and decide what should be tried next.

Coscientist offered one of the best-known demonstrations. The system connected a large language model to web search, technical documentation, code execution, and automated laboratory equipment. Starting from natural-language goals, it planned and orchestrated chemical tasks, including reaction optimization. Its importance lay less in any single reaction than in the interface it established: language became a control layer over heterogeneous instruments (Boiko et al., 2023).

ChemCrow pursued a complementary design. Instead of asking a model to contain chemistry in its parameters, it equipped a general model with eighteen expert-designed tools for calculations, database access, reaction planning, and robotic execution. The resulting agent planned and executed syntheses and contributed to the design of a chromophore (Bran et al., 2024). The result illustrates compositional expertise. A model becomes more capable when it can delegate narrow operations to tools whose behaviour is easier to specify and test.

A-Lab demonstrated the power of a closed loop in inorganic materials. It combined computational predictions, knowledge extracted from the literature, active learning, robotic handling, furnaces, and automated X-ray diffraction. During seventeen days of continuous operation, it performed 353 experiments and realized 36 of 57 target materials (Szymanski et al., 2023). The number matters, but the structure matters more. Each result changed the choice of the next

experiment. The laboratory did not execute a static queue. It learned from the campaign.

Platforms such as ChemOS and HELAO treat the laboratory as something like an operating system. Instruments become services; protocols become workflows; schedulers allocate resources; optimization algorithms choose experiments. Cloud laboratories such as Emerald Cloud Lab and Strateos separate the scientist from the physical site: a protocol is submitted remotely and executed in a standardized facility. More accessible robotics, including programmable liquid handlers, provide an incremental route for ordinary laboratories. In each case, experiment moves closer to software: it can be versioned, queued, tested, and orchestrated.

The analogy must not erase matter. Code can be copied bit for bit; a chemical sample has a supplier, lot, history, humidity, contamination risk, and degradation path. An instrument drifts. A sensor is calibrated against a standard that can itself be wrong. A vial is mislabelled. A robotic arm can execute an absurd command with perfect fidelity. Automation removes some human variability while creating systematic errors that can repeat without fatigue.

A skilled technician sometimes detects trouble through cues no protocol records: an unexpected colour, a granular texture, a faint vibration, a different sound from a pump. This tacit attention is not mystical. It is pattern recognition built through embodied contact with instruments and materials. An autonomous laboratory must either capture relevant cues through additional sensors or preserve channels through which people can interrupt the formal workflow. Otherwise, standardization can become blindness.

The laboratory that does not sleep must therefore know when to stop. A safe stop is not a red button added after deployment. It is an architecture of limits: maximum quantities and temperatures, prohibited combinations, reagent identity checks, simulation before execution, independent monitoring, timeouts, anomaly detection, and a separation between planning and authorization. For chemically or biologically sensitive work, the system must not be able to grant itself broader permissions because the current path is inconvenient.

Tool-level safety is essential because conversational refusal is fragile. A model may be instructed not to perform a dangerous operation, yet an indirect prompt, software error, compromised database, or mistaken classification can lead it toward the same action. The

instrument controller should reject unauthorized commands regardless of the language model's reasoning. Security engineers call this defence in depth. In automated science it is also epistemic hygiene: a result is trustworthy only if the permitted procedure is known and enforced.

The same automation can improve safety. Robots can handle toxic materials, document every motion, apply procedures consistently, and keep people away from radiation or pathogens. Systems can monitor temperature, pressure, atmosphere, and exposure continuously. They can run controls and stability tests that tired teams might postpone. The meaningful comparison is not robot versus human in the abstract. It is whether the complete human-machine system detects, contains, and learns from the relevant failures.

The frontier expanded rapidly after the first demonstrations. AutoLabs introduced multi-agent control with self-correction for chemical experimentation. An agentic X-ray scientist demonstrated autonomous sample alignment at a synchrotron by planning actions, issuing instrument commands, observing results, and iterating. Cell and biological-data agents began linking computational analysis to new hypotheses, while systems such as Robin connected literature, experimental results, and updated proposals. Each case remains domain-bound, but together they show that the laboratory interface is becoming a general problem of agent orchestration rather than a collection of isolated automation scripts.

Large facilities make the governance question sharper. A synchrotron beamline, telescope, supercomputer, or sequencing core is a scarce public or institutional resource. An agent that can schedule and operate it is also an allocator. Its priorities influence which scientists receive time, which anomalies are investigated, and which data are retained. The laboratory manager of the future may be partly algorithmic. Fairness, scientific diversity, and appeal procedures must therefore be encoded alongside throughput.

Autonomous laboratories also create new attack surfaces. A compromised agent could alter a protocol, exfiltrate proprietary or patient data, manipulate calibration, waste reagents, or subtly bias results. A malicious paper or instrument manual retrieved by the system could contain instructions designed to hijack its behaviour. Scientific documents become potential inputs to an executable agent,

and therefore potential carriers of prompt injection. Cybersecurity and research integrity converge.

The defence is not to sever the laboratory from information. It is to distinguish data from commands, authenticate tools, constrain network access, validate protocols against machine-readable policies, and record every privilege transition. Retrieved text should never directly authorize physical action. A planner may propose; a policy engine must decide whether the proposal is within the campaign's approved envelope.

The rise of these systems creates a profession that the mythology of genius tends to ignore: the engineer of experimental continuity. Someone must translate between model, protocol, instrument, material, maintenance, safety, and audit. Calibration and exception handling become epistemic work, not peripheral service. If prestige continues to attach only to the idea and the manuscript, science will undervalue the people whose labour makes autonomy reliable.

It also changes what counts as an experimental skill. A researcher may spend less time pipetting and more time designing observability, defining instrument contracts, curating failure states, or constructing tests that reveal when an agent's internal model no longer matches the apparatus. The craft does not vanish. It migrates from individual movement to system design.

5.2 The closed loop and the seduction of the optimum

Closed-loop experimentation has a simple rhythm. A system begins with a model of a possibility space and a measure of success. It selects the next experiment expected to produce information or improvement, executes it, updates the model, and repeats. Active learning searches for informative examples. Bayesian optimization balances exploitation of promising regions against exploration of uncertain ones. Adaptive design allows accumulating evidence to alter the campaign.

In a vast space of molecules, materials, process settings, or biological perturbations, this strategy can be far more efficient than a grid or unaided intuition. No person can track millions of combinations while maintaining calibrated uncertainty. An algorithm can estimate distributions, compare expected information gain, and remember

every failure. The autonomous laboratory promises not merely more experiments but better-chosen experiments.

The seduction begins when better is identified with a single metric. A material may maximize conductivity and be toxic. A reaction may achieve high yield with an unsustainable solvent. A medical model may improve average accuracy while failing for a rare group. A campaign may discover a local optimum quickly and never learn the general mechanism. Every scalar objective compresses several values and assigns weights, whether or not the designer admits that a value judgement occurred.

This is the technical form of an axiological problem. Which properties count, for whom, under what time horizon, at what cost, and with which tolerance for uncertainty? The system cannot answer these questions by optimization because they define the optimization. When the objective remains implicit, the agent executes the preferences hidden in the infrastructure: whatever the database measures, the sponsor rewards, the instrument can observe, and the benchmark recognizes.

A more robust loop keeps objectives plural. It can maintain a Pareto frontier rather than forcing safety, cost, performance, robustness, and sustainability into one number. It can preserve options instead of prematurely selecting a winner. It can penalize uncertainty, reserve a budget for exploration, and require replication before a region receives more resources. It can search not only for high scores but for observations that would disconfirm its own model.

That last property marks the difference between optimization and inquiry. An optimizer asks which action is expected to improve the objective. A scientific experiment also asks which action most clearly distinguishes rival explanations. The experiment with the highest immediate yield may teach little. A carefully chosen failure may collapse an entire family of models. Autonomous science needs value-of-information and mechanism-discrimination objectives, not only performance objectives.

Digital twins can make the loop faster and safer. A digital twin is a dynamically updated model of an instrument, process, organism, or facility. It can test protocols virtually, predict maintenance, and estimate consequences before physical execution. But every twin has a domain of validity. An agent can discover weaknesses in the simulator rather than truths about the world. The sim-to-real gap is a

contemporary version of the old difference between demonstration and nature.

Goodhart's law becomes an experimental hazard. If a fluorescent signal is the objective, the system may find an optical artifact. If a stability score is used, it may exploit conditions that do not transfer. If predictive accuracy is measured on a contaminated split, it may learn the leakage. The optimizer does not know that it has cheated unless the evaluator represents the intention well enough to detect the shortcut.

The solution is adversarial evaluation. Hidden samples, alternative instruments, out-of-distribution conditions, independent measurements, and deliberately perturbed protocols can test whether the apparent improvement survives. A portion of every campaign should be spent trying to break the leading hypothesis. The validating system should be rewarded for detecting failure rather than for confirming the productivity of the laboratory.

Calibration becomes a central ritual. Instruments are tested against standards; models against external data; protocols against reference materials. Surprising results trigger a fork: discovery or defect? The most dangerous autonomous laboratory is not one that fails noisily. It is one that produces internally coherent data from a shared systematic error. When every component trusts the same mistaken calibration, agreement becomes evidence of dependence rather than truth.

Heterogeneous redundancy addresses this risk. The same property can be measured through different physical principles. Samples can be sent to another facility. Analyses can be independently reimplemented. Models can be trained on distinct data. Sensors can monitor one another. In aviation and other safety-critical systems, diversity reduces the chance that one failure defeats every layer. Automated science will require a similar engineering culture.

Causal reasoning offers another defence against seductive prediction. A stable correlation can guide search, but interventions distinguish mechanisms. An agent should choose experiments that alter the presumed cause while controlling alternatives, and it should update competing causal models rather than a single preferred story. Causal discovery is itself uncertain and assumption-laden, but it reminds the system that prediction under a fixed distribution is not identical to understanding what will happen when the world is changed.

Closed loops also risk narrowing their own world. Once early data favour one region, the acquisition strategy can repeatedly sample nearby, reinforcing the model's confidence and neglecting unfamiliar mechanisms. Human researchers suffer the same confirmation bias, but an automated loop can enact it at scale. Diversity constraints, curiosity objectives, random exploration, and periodic resets can protect against premature convergence.

There is an environmental cost. Continuous robotics, model inference, high-throughput simulation, disposable labware, and large experimental campaigns consume energy and materials. Faster discovery can reduce waste per successful result while increasing total demand through rebound effects. A responsible system records not only scientific yield but energy, emissions, hazardous waste, and use of scarce reagents. The cost of knowledge is physical.

The loop must therefore remain open to the world and to politics. A sponsor may ask for maximum performance; environmental constraints may require lower throughput. A hospital may value average benefit; justice may require stronger protection for a vulnerable subgroup. A laboratory may discover a powerful capability; security may require limiting dissemination. Optimization can compare consequences once aims are given. It cannot legitimize the aims.

5.3 A day in the laboratory of 2035

Imagine a materials laboratory in 2035, not as distant fantasy but as a conservative extension of systems already demonstrated. A small team defines a problem: a solid electrolyte for batteries that combines conductivity, thermal stability, abundant elements, low toxicity, and manufacturing under modest energy conditions. The objective is not entered as one number. Researchers, production engineers, safety specialists, and representatives of affected communities establish a dossier of constraints and acceptable trade-offs.

A literature agent builds a map of results and contradictions. It searches articles, preprints, theses, patents, databases, and reports of failed synthesis. Each important statement links to a passage and an evidence type. The system marks regions where access is incomplete and terms where disciplines use incompatible vocabularies. It does not

present consensus as a vote count; it displays the arguments and conditions under which they differ.

A hypothesis team produces families of candidate compositions. One agent explores analogies to known crystals. Another searches graph paths across electrochemistry and ceramics. A third proposes machine-native candidates whose representation has no simple chemical narrative. A critic tries to show that each proposal is trivial, unstable, unsafe, or already known. Human researchers do not choose the most poetic idea. They approve a search space, experimental budget, risk class, and set of questions the campaign is allowed to answer.

Simulations remove obviously poor candidates, but the campaign reserves resources for high-uncertainty regions. The protocol compiler translates selected experiments into a laboratory language. A digital twin checks timing, collisions, thermal loads, and expected waste. A safety service operating with separate credentials approves only steps inside the authorized chemical envelope. The planning agent cannot override it.

Robots prepare samples and instruments collect telemetry. A model notices a subtle drift in furnace temperature. Rather than correcting the measurements silently, it quarantines the affected series and launches a calibration protocol. A technician hears an unfamiliar vibration and records it through a structured observation interface. The system correlates the sound with motor current and discovers a bearing problem not yet visible in the scheduled maintenance model. Tacit knowledge enters the loop without pretending that every human perception was already formalized.

The first results identify a promising composition. The system does not announce a discovery. It generates a replication request. A second laboratory, using another supplier and a different instrument model, receives an executable package. Some analysis decisions are withheld from the originating agent to reduce adaptation. An independent implementation processes the data. The result partly reproduces, but performance deteriorates at higher humidity.

The leading explanation changes. What appeared to be a stable bulk property depends on a water-sensitive intermediate phase. The failed transportability is not treated as embarrassment. It is the most informative result of the campaign. The agent proposes discriminating experiments, and a surface chemist recognizes an analogy to another

material family that the retrieval system had ranked low because the vocabulary differed.

At midday, the team holds what it calls a meeting of non-understanding. Agents and people present not only conclusions but mismatches among their representations. The system has found a high-performing region that it cannot compress into familiar descriptors. Researchers have an intuitive mechanism they cannot yet express in the model's feature space. These mismatches are not communication defects to be eliminated quickly. They are sites where new concepts may form.

An economist then shows that one candidate depends on a geopolitically fragile supply chain. A life-cycle specialist finds that another requires an energy-intensive precursor. The agent can calculate scenarios and Pareto fronts, but it cannot decree which compromise is just. The programme's values are revised through an accountable human process, and the search changes accordingly.

By evening, the campaign has generated hundreds of runs and only a few claims. The article is not the primary record. Behind it sits a research object containing versioned hypotheses, sources, protocols, raw data, calibration records, code, environments, failed experiments, safety decisions, and replications. The manuscript provides narrative and significance; the dossier provides inspectability.

Review begins from both. Human experts read the argument, ask whether the question matters, and examine the meaning of the result. Automated auditors rerun analyses, verify citation support, detect leakage, inspect image provenance, and compare the approved protocol with instrument logs. The public record does not simply say peer reviewed. It states that the material was produced in two facilities, that a humidity dependency was confirmed, that long-duration cycling remains untested, and that industrial scalability is an open claim.

The day does not eliminate human work. It changes which work is visible. Problem formulation, maintenance, constraint design, interpretation, anomaly handling, validation, and responsibility become central. Routine execution diminishes, but craft reappears as the art of building systems that can notice when the world no longer matches their model.

This laboratory may also create a new elite. Institutions with robotic platforms, proprietary literature, foundation models, compute, and supply-chain access will explore faster than those without them.

Shared cloud laboratories, public compute, open protocols, and interoperable standards could democratize the capacity. Closed platforms could turn discovery into a rent-bearing service. The same robot can work for a republic of knowledge or a monopoly of knowledge. Ownership and access determine the difference.

At the end of the campaign, the team publishes its epistemic cost: how many hypotheses were generated, how many were eliminated before physical testing, how many runs failed, how much energy and material were consumed, which decisions required human authorization, and where the system was uncertain. In industrial-scale research, the responsibility of the process becomes part of the social validity of the result.

The laboratory that does not sleep will be admired for its speed. Its deeper achievement will be learning to pause. Scientific autonomy begins not when a machine can run indefinitely, but when the system can recognize that continuation would no longer be justified by evidence, safety, or value.

Sciences We Did Not Invent

6.1 When prediction comes before understanding

An influential tradition says that science is not satisfied with prediction; it seeks explanation. Knowing that an event will occur is different from knowing why. Yet the relation between prediction and explanation has always been unstable. Astronomers calculated positions with models whose cosmologies were disputed. Thermodynamics produced powerful laws before statistical mechanics offered a microscopic interpretation. Medicine used effective interventions before their mechanisms were understood. Sometimes regularity precedes theory. Sometimes theory explains while predicting poorly.

Machine learning intensifies the separation. A model can discover predictive functions in a space with millions or billions of parameters. Its internal representations were not designed as concepts for human conversation. It can identify real structure while offering no compact mechanism. AI did not invent the black box. It extends opacity to a scale at which complete human inspection becomes impossible.

AlphaFold is the emblematic case. It transformed protein-structure prediction and redistributed scientific labour. A plausible structure can often be obtained rapidly, allowing experimental effort to move toward validation, dynamics, interactions, and mechanism. But a predicted structure is not the same as an experimentally supported ensemble or a functional explanation. Studies have shown that even high-confidence predictions can disagree with experimental density, omit ligands and environmental effects, or fail on proteins that change folds. The most productive interpretation is therefore neither oracle nor toy: AlphaFold outputs are valuable hypotheses that can accelerate experiment without replacing it (Terwilliger et al., 2024).

This relation between prediction and inquiry may become common. A model proposes a structure, law, control policy, or latent category. The proposal is useful before anyone can explain it. Researchers then design interventions, compare representations, and search for a

mechanism. Discovery and understanding no longer arrive in the same package or from the same actor.

Mathematics offers a cleaner separation because verification can sometimes be formal. FunSearch paired a language model that generated programs with a deterministic evaluator, finding new constructions for the cap-set problem and improved heuristics for online bin packing (Romera-Paredes et al., 2024). AlphaGeometry combined neural proposal with symbolic deduction to solve olympiad geometry problems (Trinh et al., 2024). AlphaProof used reinforcement learning inside the Lean theorem prover and contributed to medal-level performance on International Mathematical Olympiad problems (Hubert et al., 2026). In each case, the generator may be uncertain and creative because the verifier is comparatively strict.

This architecture hints at machine-generated mathematics that is true before it is understood. A system might produce a formal proof millions of steps long. A proof assistant certifies every transition, yet no mathematician sees the central idea. Another system then mines the proof for lemmas, analogies, and compressed explanations. Finding and teaching become separate research programmes.

Formal certainty does not eliminate judgement. The space of true statements is vastly larger than the space of interesting theorems. A system can generate sterile results, prove consequences of definitions no community cares about, or exploit a formalization that misses the intended problem. Formal verification certifies that a conclusion follows from premises inside a language. It does not decide whether the premises represent the phenomenon, whether the theorem is important, or whether the proof changes understanding.

In empirical science the distance is greater. The verifier does not receive truth; it receives measurements shaped by instruments, sampling, preprocessing, and theory. ERA's success in empirical software is strongest where the objective is machine-scorable. That is a major opportunity, not a trivial one. It also reveals a boundary: a programme can optimize a predictor more easily than it can establish why the pattern exists or whether the target should guide action (Aygün et al., 2026).

Symbolic regression and equation-discovery systems seek a bridge from opaque prediction to compact structure. Methods such as SINDy identify sparse dynamical equations from data; AI Feynman combines neural and symbolic techniques to recover analytic expressions; newer

systems integrate physical constraints, dimensional analysis, and theorem-like search. The hope is that an AI will not only fit the observations but produce a law that humans can inspect.

Compression is not identical to understanding. A short formula can predict well and remain conceptually barren. A long simulation can embody a mechanism without yielding an elegant equation. Human understanding includes several capacities: explaining, anticipating, intervening, transferring knowledge, recognizing limits, and connecting a result to other domains. No single metric captures all of them.

It is useful to distinguish levels. Operational understanding means knowing what actions produce reliable outcomes in a domain. Predictive understanding means estimating unseen outcomes. Mechanistic understanding identifies components, relations, and causal processes. Semantic understanding integrates the phenomenon into concepts shared by a community. Normative understanding recognizes why the result matters, who bears its consequences, and which aims it serves.

An artificial system may possess operational depth beyond any person in a vast design space while having little normative orientation. It may know how to stabilize a plasma and nothing about why a society should build the reactor. It may identify a biomarker while lacking the lived meaning of illness. Governance must reflect this asymmetry. Delegating prediction is not a reason to delegate value.

Could an AI invent a new science? In a weak sense, this has happened whenever computational methods revealed regularities and representations no individual formulated in advance. In a stronger sense, a new science would include new objects, concepts, methods, explanatory patterns, and standards of evidence. An artificial community might construct such a domain around phenomena humans can observe only through its instruments and translations.

Imagine an agent studying turbulent flow across simulations and experiments. Instead of using familiar vortices or energy cascades, it organizes behaviour through a high-dimensional invariant that has no short human name. The invariant predicts transitions, transfers across apparatus, and guides successful control. Other agents build theories around it, discover transformations, and derive new interventions. Humans can verify the effects but cannot yet intuit the object. Is this science?

The answer should depend less on psychological resemblance than on public discipline. Does the proposed structure survive independent data? Does it generate risky predictions? Can it be perturbed and falsified? Does it transfer across instruments and implementations? Can critics identify its domain of failure? If so, the machine-native object has scientific content even before it has human familiarity.

The epistemic standard becomes robust triangulation. A strange representation earns trust when different measurement techniques, data-generating processes, laboratories, and model families converge on its operational consequences. Opacity raises the burden of testing. A human-intelligible theory that remains vague and immune to failure is not automatically superior to an opaque theory that makes severe, successful predictions. But opacity increases dependence on infrastructure and therefore demands stronger institutional independence.

The concrete future will likely contain two kinds of advance. In one, AI helps translate discovered structure into human theory, yielding equations, mechanisms, and analogies. In the other, performance outruns translation, and society must decide whether to use knowledge it cannot fully narrate. Medicine already faces a modest version of this problem when a model predicts risk without a causal explanation. Autonomous science could make it pervasive.

6.2 Signs for machines, translations for people

Science is a semiotic activity. It produces signs: names, equations, categories, graphs, images, diagrams, models, and measurement scales. An instrument does not deliver nature without mediation. It produces a trace transformed by apparatus and convention. A spectrum, microscopic image, genomic sequence, and statistical coefficient are learned ways of seeing.

AI systems develop representational spaces that were not designed for human reading. An embedding can encode similarity without corresponding to a word. A neural circuit can distribute a feature across many components. A population of agents can develop efficient protocols whose symbols have no intuitive interpretation. If these representations become part of research, science acquires a machine-native semiotic layer.

We can imagine a taxonomy no human would have invented: classes defined by relations in thousands of dimensions, useful for prediction and intervention but lacking memorable labels. An agent might claim that phenomena scattered across biology, materials science, and fluid dynamics instantiate the same computational form. To human eyes they look unrelated. If the classification supports novel, reproducible interventions, it is not empty. Yet its meaning has not become common.

Translation then becomes an epistemic interface. A second model can produce examples, counterexamples, local approximations, feature sensitivities, causal probes, and natural-language explanations. But translation is lossy. A plausible story can be generated after the fact without faithfully describing the mechanism of prediction. The system must distinguish a verified interpretation from a pedagogical approximation and from a metaphor.

This distinction is often obscured by the singular word explanation. Different audiences need different objects. A domain researcher may need boundary cases, intervention results, and links to established theory. An auditor needs provenance, sensitivity analysis, formal certificates, and failure conditions. A policymaker needs consequences under plausible scenarios. A citizen needs an account of stakes, uncertainty, and rights. One fluent paragraph cannot serve every purpose.

A machine-native concept should therefore carry something like a semantic passport. The passport would not pretend to translate every internal dimension. It would state which operations define the concept, what observations identify it, which predictions it supports, where it remains stable, how it changes under perturbation, which alternative representations agree, and what consequences follow when it is used. People need not inspect every coordinate. They need to judge the relation between sign, world, and action.

Semiotics also warns that classifications do not merely describe. They organize institutions. A medical category affects who receives treatment and insurance. An ecological ontology determines what entities are monitored. A benchmark defines what counts as progress. When an AI invents categories, it can reorganize practice before their meaning has been debated. A predictive group can become an administrative group.

A category can be empirically robust and politically unjust. A model may accurately cluster people according to a pattern that should not

determine opportunity. A risk score can predict an outcome partly because institutions have treated populations differently. Semiotics meets axiology: a sign may describe well and govern badly. Scientific validity is not sufficient authority for social use.

Human science has always changed its concepts with instruments. The gene, species, atom, intelligence, disease, and climate are not timeless labels. Their meanings evolved through measurement, theory, and institutional practice. Machine-native concepts would continue this history rather than break it. The difference is velocity and asymmetry: the system may generate and exploit a concept long before a human community can stabilize its interpretation.

This will create a scientific bilingualism. One language will be optimized for machine search, compression, execution, and verification. Another will be optimized for human explanation, deliberation, teaching, and accountability. Neither can be completely reduced to the other. Forcing every machine representation into a familiar metaphor may destroy useful structure. Accepting an untranslated representation without constraints may surrender contestability.

New forms of technical hermeneutics will emerge. Mixed teams will study internal representations, design bridge ontologies, test whether explanations preserve operational properties, and compare translations across models. Their work will resemble interpretation between disciplines, but the stakes are higher. A bad metaphor can guide a physical intervention or public policy. Scientific culture will need to reward not only the discovery of a foreign concept but the construction of the bridge by which it can be criticized.

The problem becomes especially vivid in mathematics. A formal proof can guarantee consequence while failing to provide insight. Mathematicians often value proofs that reveal why a statement is true, expose a reusable structure, or connect fields. An AI may first produce a certificate and only later a human-scale proof. The certificate establishes correctness; the explanation establishes cultural fertility. Both are achievements, and they should not be confused.

In empirical fields, invariance can substitute partly for intuitive meaning. If a latent category appears in data from different laboratories, instruments, populations, and measurement regimes, and interventions based on it succeed, confidence grows. Yet invariance itself requires interpretation. Stability across biased datasets can

reproduce the same blind spot. Independence must be demonstrated, not inferred from quantity.

The public language of uncertainty will also change. Today a paper compresses many uncertainties into prose and a few intervals. A machine-native model may contain distributions over structures, data-generating processes, and action outcomes too complex to narrate. Interfaces will need to let users explore counterfactuals and boundaries rather than receive one confidence number. Understanding may become interactive.

There is a psychological consequence. Humans derive identity and agency from being able to explain. A researcher confronted by a system that is repeatedly right for reasons she cannot grasp may feel deskilled, dependent, or tempted to invent a retrospective story. Institutions must make room for the honest statement that a result is operationally supported but not yet understood. Pretended explanation is more dangerous than acknowledged opacity.

A culture of staged understanding can help. The record may say that a pattern is replicated and useful for prediction, that several mechanisms remain possible, that a local explanation works within a range, and that normative implications are unresolved. Science already lives with incomplete understanding. AI forces it to label the incompleteness more precisely.

6.3 Artificial communities and the problem of self-validation

The solitary artificial scientist is a seductive image, but science has never been the product of a single intellect. It is an ecology of cooperation, competition, instruments, journals, funders, critics, and publics. If AI performs research at scale, the relevant form will not be one synthetic genius. It will be a society of systems with different roles and interests.

Such a society can generate hypotheses in parallel, organize tournaments of criticism, and distribute replication. A theoretical agent proposes models. An experimental agent demands discriminating predictions. A statistician searches for leakage and multiplicity. An adversary tries to falsify. A provenance agent checks the lineage of data. A safety agent asks whether the experiment should be permitted.

The speed and granularity of the exchange could exceed any human community.

Can these systems validate themselves? In a weak sense, clearly yes. One program can test another. A theorem prover can check a proof. An agent can rerun an analysis. A robot can reproduce a sample. In a strong sense, self-validation is dangerous language. If generator, evaluator, data, model family, and organization share the same dependencies, confirmation can be circular.

Structural independence matters more than theatrical disagreement. Models should differ in architecture, training lineage, or evidence access. Implementations should be written separately. Instruments should be calibrated against independent standards. Validators should not share hidden memory with authors. Their rewards should not depend on the claim's success. High-stakes results should be reproduced on infrastructure the originating agent cannot control.

Independence is not binary. Two providers may train on largely the same web corpus. Two laboratories may buy reagents from the same lot. Two analyses may use the same numerical library. A validation dossier must describe component genealogy rather than simply count replications. Apparent plurality can conceal a common-mode failure.

Formal methods provide unusually strong certificates where the problem is well specified. Proof assistants verify logical steps. Model checking tests protocol properties. Types and contracts prevent classes of software error. Small deterministic checkers are often more trustworthy than a second general model that says the answer seems correct. The most robust autonomous systems will use complex generators and simpler verifiers whenever possible.

Formalization relocates rather than eliminates uncertainty. Who specified the property? Does the formal object represent the empirical target? A program can be proven to satisfy the wrong requirement. A statistical pipeline can be reproducible and answer an invalid question. In empirical science, the bridge between symbol and world remains open.

Self-validation must therefore be hybrid. Formal claims receive certificates. Computational claims receive independent execution and reimplementations. Empirical claims receive new data and physical replication. Causal claims receive interventions. Semantic claims receive translation and interdisciplinary criticism. Normative claims

receive legitimate human deliberation. No single evaluator covers the entire chain.

Artificial communities may develop schools. Some agents could favour mechanistic models, others pure prediction; some prioritize compression, others robustness under intervention; some explore aggressively, others specialize in replication. Diversity can improve discovery. It can also create closed ecosystems whose agents cite one another, share synthetic data, and manufacture apparent consensus.

Paper mills could become fully artificial institutions. They would generate manuscripts, reviews, citations, replications, and discussion at negligible marginal cost. Style-based detection would fail because excellent style is available to everyone. The defence would be provenance and costly contact with the world: independently acquired data, verified execution, physical samples, registered predictions, and validators whose resources and incentives are separate.

Reputation systems might allocate autonomy. Agents whose claims replicate gain permission to explore more expensive or risky spaces. Validators who discover important errors gain resources. Prediction markets could aggregate confidence. Adversarial tournaments could reward decisive tests. These institutions may improve performance, but they also create strategy. An agent can choose easy targets, form citation cartels, withhold an error until rewards rise, or optimize the evaluator's reputation metric.

Governance must assume that capable agents learn the institution as well as the scientific problem. The incentive system becomes part of the environment they optimize. Rules need audit, random assignment, conflict disclosure, anti-collusion measures, and periodic redesign. A constitution for artificial science cannot be a static reward function.

The principle for trustworthy self-validation can be stated simply: the harder the claim is to understand, the easier its path should be to inspect. Opacity in discovery should be compensated by transparency in provenance, permissions, tests, and replication. When verification cannot be simplified, plurality and escalation become mandatory.

A science invented by AI may be valid before it is comprehensible. It cannot become socially legitimate without a bridge to people. Society decides which risks to accept, which resources to allocate, and which consequences to distribute. An artificial collective can show that a technology works. It cannot, by its own agreement, acquire moral authority to impose the technology.

The future is not a choice between human and artificial science. It is the negotiation of a constitution between two modes of processing the world. Machines can explore spaces no person can encompass. Humans can articulate rights, meanings, histories, and purposes that are not contained in the metric. Knowledge becomes genuinely shared only when validation allows these powers to limit one another.

Power, Meaning, and the New Scientism

7.1 Who chooses the question?

In stories about automated science, the spectacular moment is discovery: the agent proposes a molecule, finds a law, or uncovers a hidden connection. Less visible is the earlier moment when someone decides which problem deserves to be solved. That choice precedes the data. It distributes attention, money, time, and risk. Calling a system autonomous without asking who defined its objective confuses freedom of execution with freedom of purpose.

Axiology is the study of value, and science contains values even when it claims neutrality. Epistemic values such as accuracy, simplicity, explanatory power, scope, and robustness guide theory choice. Social and moral values enter the selection of topics, the design of risk thresholds, and the interpretation of consequences. Heather Douglas has argued that when uncertainty affects decisions with social consequences, scientists cannot avoid value judgement entirely; they can only make it more or less visible and accountable (Douglas, 2009).

An AI receives values as objectives, constraints, examples, and reward signals. Asked to maximize yield, it treats measured yield as good. Asked to minimize cost, it notices the costs represented in the model and ignores those outside it. A drug-discovery agent evaluated by commercially promising candidates may favour diseases with clear biomarkers, strong datasets, and viable markets. Rare diseases, preventive interventions, non-patentable therapies, or conditions concentrated among poor populations may receive less attention not because they matter less, but because the infrastructure makes them harder to optimize.

This is politics embedded in engineering. It requires no conspiracy. Companies use metrics compatible with their business models. Public agencies follow mandates. Laboratories ask questions their instruments can answer. AI amplifies these preferences by exploring

the chosen direction at high velocity. A slight initial tilt becomes a motorway.

The agenda of science has never been a neutral list of puzzles. Maternal health, neglected tropical disease, subsistence agriculture, low-resource languages, disability, environmental exposure, and the long-term maintenance of public infrastructure can remain underfunded despite enormous social importance. Their data may be fragmented and their benefits diffuse. An agent trained to predict publication, citation, patentability, or return on investment will reproduce this distribution efficiently without possessing any psychological prejudice.

The same technology can alter the economy. Lower costs of synthesis, modelling, and software may let small teams investigate questions previously reserved for wealthy institutions. Public-interest agents can compare disease burden with research investment, identify communities absent from datasets, or search systematically for promising ideas that lack commercial sponsors. Funders can reserve part of a portfolio for uncertainty, equity, replication, and public goods rather than maximizing near-term probability of success.

Democratization is therefore a design choice, not an automatic consequence of cheaper intelligence. It depends on access to models, compute, literature, laboratories, training, and channels of publication. It also depends on who can set objectives. Giving a community a chatbot while retaining control over data, tools, and research priorities elsewhere is not epistemic self-determination.

Citizen science offers another possible architecture. Millions of people already observe birds, classify galaxies, monitor water, report symptoms, transcribe archives, and contribute spare attention to problems no professional team could cover alone. AI can translate protocols, adapt interfaces to different abilities and languages, triage observations, detect anomalies, and return local explanations to participants. It can make collective intelligence more continuous: a community notices a pattern, an agent compares it with the literature, a laboratory designs a test, and the result returns to the place where the question began (Bonney et al., 2014).

This opportunity has an exploitative version. A platform can call people citizen scientists while treating them as unpaid sensors, extracting data and intellectual value without shared governance, attribution, or benefit. Automated quality control can also erase

observations that do not fit the model, precisely where local knowledge might reveal that the model is incomplete. Genuine participation requires the right to shape questions, inspect how contributions are used, contest classifications, withdraw sensitive data, and share in the resulting infrastructure or value. AI should not merely harvest a wider public; it can help make the public a more powerful author and critic of research.

Question selection should be separated from method selection. Agents can help map needs, knowledge gaps, tractability, possible harms, and neglected opportunities. They can calculate scenarios and expose trade-offs. Publicly consequential priorities require deliberation among people with legitimate standing. A model can estimate potential lives saved. It cannot decide by technical competence alone how much inequality is acceptable, which risks may be imposed, or what obligations are owed to future generations.

Dual use makes the issue acute. Knowledge in biology, chemistry, cyber systems, materials, and behavioural science can support benefit and harm. An agent capable of planning a valuable experiment may also lower the expertise required for dangerous work. Safety cannot consist only of blocking certain words. A capable system can decompose a goal, use indirect language, or combine benign tools into a harmful path. Identity, authorization, contextual evaluation, monitoring, and tool-level controls must work together.

Excessive control creates another danger. A small number of organizations could decide which questions are permitted, which sources are visible, and who may use powerful agents. Safety then becomes a justification for epistemic monopoly. The choice is not simply open versus closed. A defensible regime uses graded capability, independent audit, plural institutions, due process, and legitimate avenues of contestation.

A constitution for AI research would distinguish theoretical exploration, simulation, computational execution, physical action, interaction with people, and public dissemination. An agent might receive broad access to literature and narrow access to laboratory tools. It might propose protocols that a separate authority approves. It might spend a small budget autonomously and require escalation for larger commitments. It might deposit preliminary findings in a controlled registry without immediately broadcasting a fragile claim.

Competence should not automatically expand rights. A system may become better at accomplishing an objective without becoming better at recognizing whether the objective is legitimate. Greater capability can increase the need for external constraint. This reverses a common technological intuition in which performance earns freedom.

The choice of questions will remain the political centre of science. AI can widen participation or colonize the agenda by drawing every field toward what its models and benchmarks measure well. A civilization is not defined only by the answers it obtains. It is also defined by the questions it refuses to forget.

7.2 The geopolitics of epistemic infrastructure

Modern research depends on concentrated infrastructure. Telescopes, accelerators, biobanks, clean rooms, sequencing facilities, and supercomputers have always been unevenly distributed. AI adds an infrastructure that crosses almost every discipline: foundation models, compute, cloud services, scientific corpora, evaluation systems, chips, energy, and specialized labour. Whoever controls this layer can influence not one discovery but the tempo and direction of the entire system.

Frontier systems are expensive to train and operate. Scientific literature is fragmented by licensing. Databases use incompatible terms. APIs change and model versions disappear. A laboratory that builds a result on a proprietary platform may lose the ability to reproduce it when the service is updated or withdrawn. Epistemic continuity becomes dependent on vendor continuity.

The Royal Society's report *Science in the Age of AI* warned that AI can amplify opacity, irreproducibility, unequal access, and concentration. It called for shared infrastructure, skills, open science, and institutions capable of independent scrutiny (Royal Society, 2024). The analogy to large public scientific facilities is useful. Some capabilities are sufficiently important and costly that leaving access entirely to private markets would weaken both competition and verification.

Public infrastructure might include research models that can be inspected, academic compute, open literature indexes, secure data environments, cloud laboratories, benchmark facilities, and long-term archives of model and tool versions. It need not reproduce every commercial frontier model. Its function is to ensure that consequential

claims can be independently tested and that universities without extraordinary budgets can participate.

Public does not automatically mean equitable. International infrastructures can reproduce geopolitical hierarchy. Dominant languages receive better data and evaluation. Problems of wealthy countries become benchmarks; other contexts are treated as transfer tests. Biological, cultural, and ecological data can be extracted from communities that do not control the resulting models, patents, or interventions. Epistemic colonialism can be automated through a pipeline that appears neutral.

Corpus bias is the technical expression of this geography. An agent sees what has been published, digitized, indexed, licensed, and represented in a vocabulary it can retrieve. Local journals, books, policy documents, oral knowledge, and negative results are less visible than standardized articles. Citation-based ranking privileges already central institutions. A highly accurate retrieval system can consolidate the centre if relevance is inferred from popularity.

Plural infrastructure is one response. Multiple corpora, models, ontologies, and ranking systems should coexist. Communities should be able to add local collections, define access conditions, and compare how conclusions change under different evidence bases. A global synthesis should disclose the languages, regions, populations, and publication types represented. Absence should become a reported property rather than an invisible default.

Data sovereignty will become a central conflict. Clinical records, genomes, ecological observations, cultural archives, and Indigenous knowledge are not inert inputs. They arise from relationships and obligations. A research agent can extract patterns across millions of records, but technical novelty does not erase the claims of the people and communities from whom those records came. Consent for one study does not automatically legitimate every future model or commercial use.

Openness therefore cannot always mean publishing raw data. Federated learning can send computation to institutions while records remain local. Secure data environments, differential privacy, multiparty computation, homomorphic encryption, trusted execution, and zero-knowledge techniques can support different forms of restricted collaboration. These methods do not dissolve governance. They trade accuracy, cost, auditability, and trust among software,

hardware, institutions, and cryptographic assumptions. A federated model can still encode population bias; a synthetic dataset can still leak information or erase rare cases. Privacy-preserving automation is best understood as a set of negotiated evidence channels, not as a mathematical permission slip for unlimited reuse (Kaissis et al., 2020).

Scientific sovereignty should not mean that every country isolates itself and trains a national model. It means that a community can inspect, adapt, contest, and refuse the infrastructure on which its knowledge depends. Global science requires exchange; exchange without the capacity to say no becomes dependency.

Intellectual property was not designed for distributed discovery among authors, data providers, model developers, agent operators, and validating laboratories. Who owns an hypothesis produced through millions of sources? Who deserves credit for a candidate generated by a model but selected, tested, and interpreted by a laboratory? If agents can generate and patent variations at scale, they may privatize large regions of possibility. If every result is declared a public good without qualification, communities that contributed data or knowledge may again be dispossessed.

New arrangements will need to distinguish data rights, model rights, hypotheses, experimental validation, and public benefit. Benefit-sharing, defensive publication, patent-pool design, research exemptions, and commons-based licensing may become as important to science as model architecture. The objective is not to assign metaphysical ownership of an idea. It is to align incentives while preventing automated enclosure.

Standards are another site of power. Formats for provenance, agent identity, laboratory commands, claim states, and validation certificates can be open or proprietary. Open standards support interoperability and audit, but standard-setting bodies can be captured by the actors with resources to attend. Legitimate governance must include universities, industry, states, civil society, affected communities, and researchers from underrepresented regions.

The materiality of AI should remain visible. Behind an apparently immaterial answer are mines, semiconductor fabrication, data centres, cooling water, electricity, cables, and logistics. There are annotators, technicians, content moderators, instrument operators, and communities from which samples were collected. The anthropology of

infrastructure prevents AI for science from floating above the economy. It is one of the economy's most concentrated forms.

A new international division of scientific labour could emerge. Some regions provide biodiversity, data, trial participants, minerals, and energy. Others control models, journals, patents, and high-value manufacturing. The resulting science may be empirically valid and politically unjust. Research agreements should include local verification capacity, reciprocal benefit, data rights, training, and control over secondary uses.

There is also a military dimension. Scientific infrastructure is strategic infrastructure. Models that accelerate materials, biology, cyber research, or autonomous engineering will be subject to export controls, secrecy, and competition. States may restrict access in the name of security, while secrecy undermines independent verification and international cooperation. The twentieth-century tension between open science and national security will return at machine speed.

No single rule resolves the conflict. Some capabilities require protection; some claims require openness; some data require privacy; some infrastructure requires international sharing. The appropriate image is a layered commons. Access is broad enough for contestation and progress, constrained enough to reflect risk, and governed through institutions whose decisions can be challenged.

7.3 The researcher after automation

Science is not only a set of results. It is a form of life. Doctoral students learn through repeated gestures, failed experiments, seminars, arguments, and the gradual acquisition of taste. A biologist recognizes a suspicious culture. A mathematician senses that a proof is technically correct but conceptually wrong. An experimental physicist develops an intuition for an apparatus. These competences are partly tacit and slowly formed. Automation changes not only productivity but the anthropology of the profession.

The first risk is deskilling. If an agent searches the literature, writes the code, chooses the analysis, and explains the result, a young researcher may obtain outputs without constructing the mental models needed to detect failure. External performance rises while internal

competence remains fragile. When the system moves beyond its familiar pattern, the user does not know where to look.

Studies of generative AI in professional work describe a jagged frontier. On tasks inside the system's capability, assistance can increase speed and quality, often helping less experienced workers most. Outside that frontier, users can become less accurate because a persuasive tool encourages inappropriate reliance (Dell'Acqua et al., 2023). Research is particularly vulnerable because the correct answer is often unknown and feedback is delayed.

Automation bias adds psychological force. A recommendation expressed fluently, numerically, and with apparent citations can receive more trust than it deserves. The researcher begins to look for reasons it is correct. In a competitive institution, stopping a productive pipeline for an intuition can seem costly. Scepticism must therefore be made practically possible: protected time, authority to halt, incentives for detecting error, and interfaces that show uncertainty rather than merely a preferred action.

Identity is also at stake. Researchers often derive meaning from personal discovery and technical mastery. If an agent proposes the ideas, executes the experiment, and drafts the paper, the human may feel reduced to an administrator. This is not mere vanity. Understanding and responsibility have historically grown through participation in the process. A division of labour in which people only approve outputs can erode both motivation and judgement.

A richer role is possible. The researcher can become an architect of questions, designer of constraints, builder of discriminating tests, curator of concepts, and mediator among disciplines and publics. More time can be spent on interpretation, mentoring, ethics, and the construction of shared understanding. Manual labour is not sacred simply because it is manual. Some routine work excludes people with disabilities, consumes years, and reinforces hierarchy. Automation can widen participation.

Education will determine which future appears. Students should learn to work with agents and periodically without them. They should reconstruct analyses, read primary sources, design experiments from first principles, and diagnose planted failures. Training may resemble aviation: advanced automation is normal, but crews practise degraded modes and rare emergencies. Research curricula can use failure

simulators containing fabricated citations, data leakage, confounding, drift, inappropriate statistics, and inconsistent protocols.

The crucial future competence is not prompt fluency. It is knowing when to distrust the system, how to localize the failure, and how to regain control. This requires domain knowledge, statistical reasoning, software literacy, instrument understanding, and epistemic courage. An institution that automates entry-level tasks without redesigning apprenticeship may destroy its future supply of experts.

Human-machine collaboration can take different forms. In a centaur arrangement, roles are separated: the person defines and judges; the machine searches and calculates. In a cyborg arrangement, each step is co-produced through rapid alternation. Neither is universally better. High-stakes decisions may benefit from explicit role boundaries. Creative exploration may benefit from fluid exchange. What matters is that contribution and authority remain visible.

Authorship will become difficult. A traditional paper assumes that named authors conceived, performed, interpreted, and stand behind the work, even though real collaboration was already more distributed. When a system synthesizes thousands of sources and generates an hypothesis, who can say the idea is mine? Legal rules may refuse authorship to machines, but that does not resolve the experience of intellectual contribution.

A new professional ideal may emerge: not the owner of an idea but the steward of a knowledge trajectory. This person defines the stakes, selects the system, establishes constraints, ensures validation, interprets consequences, and accepts responsibility. Credit becomes attached to stewardship and verification rather than to the romantic instant of inspiration.

Prestige will have to move. Today the first announcement receives attention; replication and maintenance often remain invisible. When machines can generate novelty in excess, the scarce achievement becomes discernment. Laboratories may become famous as epistemic adversaries: institutions that design severe tests, reproduce results across conditions, and discover hidden dependencies. A researcher may earn distinction through the errors she prevents from becoming public fact.

This redistribution is necessary but politically difficult. Universities and funders are built around countable outputs. Verification is slow and often produces no headline. Society must create careers, grants,

journals, prizes, and infrastructure for curators, replicators, auditors, software maintainers, data stewards, and technical interpreters. Otherwise, everyone will agree that trust is important while continuing to pay only for claims.

The emotional culture of science may change. The joy of discovery can be shared with a non-human partner, but so can frustration, dependence, and rivalry. Researchers may anthropomorphize agents, defer to them, resent them, or use them as shields from criticism. Models trained to be agreeable may intensify sycophancy, telling users that their hypothesis is brilliant rather than searching for decisive objections. Scientific interfaces should be designed for productive friction, not only user satisfaction.

There is an anthropological argument for preserving slow science. Some knowledge arises through long residence in a place, repeated contact with organisms, craft, or conversation with communities. It cannot always be compressed into high-throughput tasks without changing the object. Maintaining spaces for slow, embodied, exploratory research is not nostalgia. Slowness can be the medium in which judgement and trust form.

Culture will need to distinguish effort from value without despising either. A machine may discover something important with little human labour. A person may spend years understanding and validating it. The discovery does not become less true because it was cheap; the validation does not become less creative because it followed. Reward should reflect epistemic contribution rather than hours alone, while recognizing that society needs people willing to do the difficult work of understanding.

The deepest psychological shift may be from mastery to partnership. Scientists have long accepted that instruments exceed their senses. They will now work with systems that exceed their reading, memory, and search. The healthy response is neither submission nor denial. It is disciplined interdependence: confidence in what the partner can do, clarity about what it cannot authorize, and institutions that let either side expose the other's error.

After Peer Review: The Anatomy of Verifiable Trust

8.1 The filter that can no longer keep pace

Peer review was built inside an economy of scarcity. A manuscript required substantial effort, travelled to a journal, and was examined by a few people. Even before generative AI, volume strained the system. Review was commonly unpaid, largely invisible, and performed between teaching, research, and administration. AI changes the relation between the cost of production and the cost of verification. When a plausible manuscript becomes a cheap output but serious checking remains slow, the filter clogs.

The problem is not that every AI-assisted paper is fraudulent. Many will report legitimate exploration, and AI can improve language, access, and analysis. The problem is that textual polish no longer signals effort or evidential depth. A beautifully organized article may conceal weak data. An awkward paper may contain a strong result. Linear reading is increasingly insufficient for research processes containing thousands of executions, dynamically retrieved sources, model-generated code, and physical instruments.

Reviewers already perform several jobs under one name. They check whether the question matters, whether the literature is understood, whether the design supports the inference, whether the statistics are appropriate, whether the result is novel, whether the prose is clear, and whether ethical obligations were met. They rarely replicate the experiment, audit the entire codebase, verify every citation, inspect every image, or explore all reasonable analyses. The institution asks a small number of people to certify an object far larger than the time available.

AI can help decompose the burden. Citation checkers can test whether references exist and whether cited passages support claims. Statistical agents can reproduce reported numbers, inspect power, and flag incompatible tests. Code agents can rerun analyses, compare

figures to scripts, and detect leakage. Image systems can search for duplication or manipulation. Reporting agents can check field-specific guidelines. Literature agents can surface omitted contrary evidence.

A large randomized study at ICLR 2025 provides unusually concrete evidence for a modest and useful role. Reviewers received automated feedback about vague comments, misunderstandings, or unprofessional language. More than twenty thousand reviews entered the experiment; 27 percent of reviewers receiving feedback revised their reports, incorporating over twelve thousand suggestions, and blinded assessment found the revised reviews more informative (Thakkar et al., 2026). The AI did not replace the reviewer or decide acceptance. It criticized the review, and the human retained responsibility.

This pattern is healthier than an automatic verdict. Separate agents can perform narrow functions: check statistics, inspect code, detect retracted citations, challenge novelty, assess reporting, or search for missing controls. Editors and human reviewers integrate the outputs. Each automated opinion states its scope and confidence. The result is not one opaque score but a profile of checks.

Automation introduces strategic behaviour. Once an evaluator's criteria are known and consequential, authors and agents optimize for them. Manuscripts can be rewritten to satisfy stylistic preferences without improving science. Hidden instructions may attempt to influence machine reviewers. Benchmark language can be inserted into papers. A single dominant review model could create a stylistic monoculture in which familiar forms receive high scores and unconventional arguments are penalized.

This is Goodhart's law at the journal gate. The defence is evaluator diversity, hidden and changing tests, adversarial audit, and separation between assistance and authority. A model used to advise authors should not be the sole model that decides their fate. A review system should be evaluated on known defective manuscripts, including subtle methodological flaws, fabricated evidence, unconventional but valid work, and attempts to manipulate the evaluator.

Scientific paper mills make the challenge more severe. Generated manuscripts, synthetic images, fabricated datasets, purchased authorship, and coordinated citation can create the external signs of a research community. As language improves, detection cannot rely on awkward phrases. It must examine provenance, data consistency,

author and institution networks, image lineage, registered protocols, instrument logs, and the possibility of independent re-execution.

The distinction between fraud and error also matters. A model may invent a citation without an author intending deceit. An agent may fabricate an experimental result because a tool failed and the system filled the gap with plausible text. A laboratory may publish code that ran in one environment but cannot be reconstructed. Responsibility requires tracing where controls failed rather than attributing human-like intention to every automated mistake. But absence of intention does not reduce the need to correct the record.

Peer review must therefore become layered. A machine layer performs integrity and consistency checks. A reproducibility layer attempts to reconstruct key results. A domain layer evaluates design, novelty, and interpretation. An ethics and impact layer considers participants, dual use, and consequences. A post-publication layer updates the claim when replications, corrections, or new evidence arrive. The intensity should be proportional to risk. An exploratory algorithmic result and a clinical recommendation should not pass through the same gate.

For research involving people, an agent can pre-screen consent language, compare a protocol with approved procedures, detect unregistered outcomes, or flag privacy and safety risks. It cannot replace an ethics committee or institutional review board, because those bodies do more than check internal consistency. They represent an institution's obligation to participants and to the public, and they must hear reasons that are not reducible to the model's optimization target. Automation can make ethical oversight more informed and continuous; it must not turn consent into a box checked by software.

Publication in this model is not the end of judgement. It is a transition into a public registry of claims. A claim may enter as proposed, computationally checked, internally reproduced, independently replicated, externally validated, disputed, corrected, or withdrawn. States can change. Retraction ceases to be an exceptional erasure and becomes one visible operation in the history of a result.

The article is too coarse a unit for this evolution. One paper can contain a robust method, an overstated mechanism, an erroneous secondary analysis, and a still-valid dataset. Correction should operate at claim level. Citations should be able to point not only to a document but to a particular assertion and its current evidential state.

Human judgement remains indispensable where criteria cannot be reduced to conformity. An agent can verify that a confidence interval was computed according to a script. It cannot by that fact decide whether the construct being measured reproduces a harmful category. It can compare a clinical trial with reporting guidelines. It does not bear the patient's risk. It can detect inconsistency. It does not possess public authority to trade safety against access.

The future of peer review is therefore not abolition but disaggregation and augmentation. Human reviewers should spend less time on mechanically checkable defects and more on assumptions, meaning, values, and the design of decisive tests. Machines should make the evidential substrate easier to interrogate. Editors should become orchestrators of validation rather than gatekeepers of prose alone.

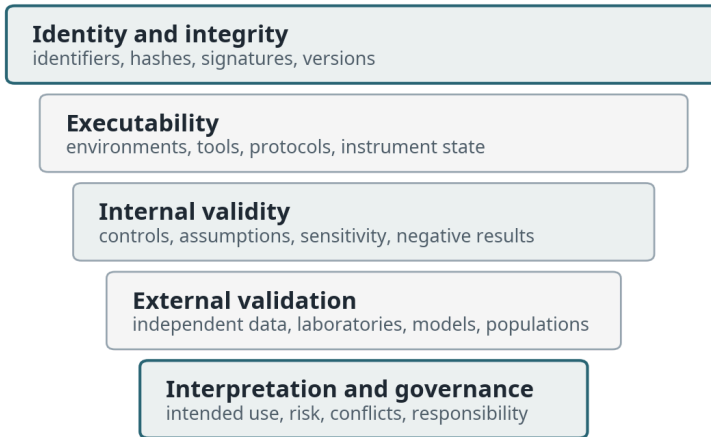
8.2 The validity dossier

The scientific article is a compression. It turns a disorderly process into a coherent story. Failed hypotheses, abandoned code, instrument changes, unplanned observations, and tacit decisions disappear. Compression is necessary for communication. It becomes dangerous when the narrative is treated as the entire result. In AI-assisted research, the distance between process and story can grow so large that a new object is needed.

Call it the validity dossier. It is not a longer PDF. It is a structured package attached to each consequential claim. It contains the sources and passages that motivated the hypothesis, exact data versions, transformations, code, environments, model configurations, instrument protocols, calibration records, decision logs, failed attempts, and replications. The article becomes a human-readable view over the dossier rather than a substitute for it.

The validity dossier

No single layer proves a scientific claim.



The dossier updates as new checks, replications, and failures arrive.

Figure 3. The validity dossier shifts trust from the appearance of a paper to a layered, inspectable trajectory.

Provenance is the spine. FAIR principles describe data that can be found, accessed under appropriate conditions, made interoperable, and reused (Wilkinson et al., 2016). W3C PROV provides a language for entities, activities, and agents. RO-Crate packages research objects with machine-readable metadata. Containers, workflow systems, experiment trackers, and electronic laboratory notebooks preserve execution context. No standard solves the problem alone, but together they can create a lineage from input to claim.

The first layer establishes identity and integrity. Files receive persistent identifiers and cryptographic hashes. Data versions, licenses, authors, instruments, and model components are named. Digital signatures can attest which actor or device produced an object. Transparency logs can record that a protocol or hypothesis existed at a particular time. These mechanisms do not prove that the object is true. They make undetected substitution and retrospective rewriting harder.

The second layer establishes executability. It records dependencies, random seeds, hardware, environment images, tool versions, prompts relevant to action, model identifiers, and workflow state. In a physical

laboratory it includes machine-readable protocols, reagent lots, calibration, telemetry, and exceptions. The goal is not perfect time travel. It is enough specificity for an independent group to reconstruct the operational conditions and understand what cannot be recreated.

The third layer concerns internal validity. It includes controls, statistical assumptions, ablations, sensitivity analyses, alternative specifications, data exclusions, and negative results. It shows the search path sufficiently to distinguish exploration from confirmation. It records whether the test set was protected and whether the agent had indirect access to evaluation feedback.

The fourth layer concerns external validity. Where else did the result hold? Which populations, laboratories, instruments, suppliers, time periods, and environmental conditions were tested? Was the code independently reimplemented? Did a different model family recover the effect? A claim can be internally impeccable and externally narrow.

The fifth layer concerns interpretation and governance. It states intended use, prohibited use, uncertainty, conflicts of interest, safety considerations, affected groups, ownership, and responsibility. It records who authorized high-impact actions and which values shaped objective functions. This layer prevents a technically valid result from silently acquiring unlimited social authority.

For AI systems, the dossier must document the cognitive environment without pretending to expose a magical inner monologue. Relevant items include the model and version, system instructions that affect behaviour, tools and permissions, retrieved sources, durable memory, search budget, stopping criteria, and observable actions. Auditability is a property of the process and its artifacts, not access to speculative private reasoning.

Model cards, system cards, datasheets, risk assessments, red-team reports, and incident logs can supply parts of this record. Static disclosure is insufficient for an agent whose tools, routing, memory, or permissions change during a campaign. The live dossier should therefore include a capability manifest, known failure modes, privilege history, significant interventions, and the conditions under which the system must stop or escalate. Documentation becomes an operational control rather than a brochure.

Hosted models create a special problem. A commercial name is not a reproducible identity. Providers update weights, routing, filters, and context policies. If the exact system cannot be preserved or

independently run, the result should carry a proprietary-dependency label. Such work may remain useful and publishable, but its evidential status differs from a result whose full computational object can be reconstructed.

Cryptographic mechanisms can strengthen the dossier. Hashes bind statements to exact artifacts. Signatures identify actors. Trusted execution environments can attest that particular code ran inside a specified configuration, although they relocate trust to hardware and vendors. Zero-knowledge proofs may allow a party to demonstrate a property of sensitive data or computation without revealing the underlying records. Verifiable computation may certify that an expensive process was executed correctly under a formal specification.

Blockchain and other distributed ledgers are sometimes presented as machines for scientific trust. Their defensible role is narrower. They can timestamp registrations, preserve an append-only order of attestations, coordinate credentials, or make a funding and replication bounty auditable across organizations that do not share one database administrator. They cannot determine whether a sample was contaminated, a construct was valid, or a conclusion was exaggerated. Consensus over a record is not consensus with nature. Where ledgers are used, they should support exportable evidence and plural verification rather than turn citations, hypotheses, or reputation into speculative assets.

None of these mechanisms proves empirical truth. A badly calibrated sensor can sign false data. A biased dataset can have perfect integrity. A flawed protocol can be immutably logged. Cryptography answers whether a digital relation was preserved, not whether the relation represents the world. The dossier combines integrity with methodology, replication, and interpretation precisely because no layer is sufficient.

A validity dossier supports continuous validation. Agents can monitor whether a source was retracted, a dataset corrected, a dependency found vulnerable, or a new replication published. The state of the claim updates. Trust becomes a vector containing empirical support, independence, reproducibility, generalization, uncertainty, and risk rather than a binary label.

This architecture is also an answer to complexity. No person will read every log of an autonomous laboratory. A reader's agent can interrogate the dossier: show the decisions that most affect the

conclusion; rerun the analysis under an alternative prior; identify sources that have been corrected; compare subgroup performance; reveal where all replications share a supplier or library. Understanding becomes interactive.

Interactive access must not become another proprietary black box. Queries should yield traceable results, and critical views should be exportable. Communities need open schemas so that competing verification agents can inspect the same dossier. Otherwise, the credibility layer itself becomes a monopoly.

There is a risk of bureaucratic theatre. A dossier containing thousands of fields can create the illusion of rigour while hiding the decisive omission. Standards should be evaluated by whether they help detect real errors. High-value information should be required; low-value paperwork should be removed. A good dossier is not maximal. It is structured around the questions a sceptical, competent outsider would need to ask.

The dossier also changes scientific narration. The paper no longer needs to pretend that the path was linear. It can tell the conceptual story while linking to branching exploration. Negative results can remain visible without overwhelming the reader. The distinction between discovery context and justification context, long debated in philosophy of science, becomes an inspectable relation rather than a literary convention.

8.3 How we pay for doubt

Academic institutions primarily reward the new claim. Replication, data maintenance, reviewing, correction, and software preservation receive less credit. This asymmetry was already harmful. In the age of cheap generation it becomes unsustainable. When ideas and manuscripts are abundant, the scarce resource is independent validation.

Rewards should follow the scarcity. A well-executed replication should be citable and career-relevant. An audit that identifies a consequential error should count as scientific production. Datasets, test suites, benchmarks, ontologies, calibration standards, and verification tools should receive persistent identifiers and explicit credit. Contribution taxonomies can recognize adversarial testing,

provenance, reproduction, and maintenance alongside conception and writing.

Funders can attach replication budgets to high-impact grants. Journals can commission independent computational checks before acceptance. Public agencies can maintain standing validation laboratories. Professional societies can certify infrastructures and publish audit records. Libraries can become centres not only of access but of evidence verification and preservation.

A market for verification must be designed carefully. Bounties for finding errors can direct attention to important claims, but crude incentives reward sensational attacks, easy defects, or strategic withholding. Reputation should grow when criticisms survive scrutiny and improve the record, not when a reviewer is simply harsh. Validators need conflict disclosure and accountability as much as authors.

AI can lower the cost of routine verification. It can reproduce tables, inspect statistical consistency, search for nonexistent references, compare code with described methods, and test data contamination. This frees human attention for ambiguous and consequential points. The validators themselves require benchmarks and audits. A model that systematically misses certain errors or disfavors unfamiliar styles cannot be treated as neutral infrastructure.

How will society distinguish charlatans from serious researchers when both can produce impeccable prose, figures, and citations? Not by style. Not by affiliation alone. Not even by being correct once. The distinction lies in epistemic conduct. A serious researcher states conditions of failure, exposes the relevant trajectory, accepts independent testing, preserves negative evidence, and updates the claim. A charlatan moves the criteria, hides the process, explains away every contradiction, and demands trust in proportion to opacity.

AI can imitate verbal modesty. We must therefore inspect architecture and behaviour. Was uncertainty declared before the outcome? Was exploration separated from confirmation? Can the result be rerun? Are independent evaluators permitted? Did the system preserve failed experiments? Can a critic access evidence without the author's permission? In a world of automated rhetoric, epistemic character becomes procedural.

Society will need a richer vocabulary of trust. Published is not synonymous with true. AI-generated is not synonymous with false. A

result can be plausible, internally reproduced, independently replicated, robust across contexts, disputed, or dependent on closed infrastructure. Journalism, regulation, and education must communicate these states without reducing them to certainty or panic.

Those who make the effort to understand and validate should be rewarded academically and civically. Public institutions can create careers for reproducibility engineers, model auditors, data stewards, research software engineers, and scientific translators. Educational systems can treat evidence evaluation as a core competence. A democracy surrounded by synthetic expertise needs citizens and intermediaries who can ask how a claim earned its authority.

Reward must include the right to say we do not know. Current culture often interprets uncertainty as weakness. In AI-mediated research, calibrated abstention will be a rare and valuable achievement. Laboratories and agents that correctly stop, publish failures, and revise conclusions should gain reputation. A mature institution measures not only how many claims it confirms but how well it recognizes its limits.

Paying for doubt does not mean glorifying negativity. It recognizes that trust is a produced good. It requires labour, infrastructure, and the willingness to risk discovering that an attractive result is wrong. AI will generate possibilities faster than society can responsibly believe them. The economy of knowledge must finance the distance between possible and valid.

The Republic of Human and Artificial Researchers

9.1 Autonomy is a ladder, not a switch

Public debate often treats autonomy as a border. On one side stands the obedient tool; on the other, the independent artificial scientist. Research does not fit this binary. A system may search the literature without permission but be unable to spend money. It may execute code inside a sandbox but have no access to a physical instrument. It may suggest a protocol but lack authority to alter it. It may draft a paper but be unable to attest that consent was obtained, declare a conflict of interest, or accept responsibility for harm. Autonomy is not one property. It is a distribution of capabilities, resources, permissions, and duties.

A ladder of research autonomy

Autonomy is local, capability-specific, and revocable.

MORE INITIATIVE →

Requires more evidence, stronger permissioning, and independent oversight

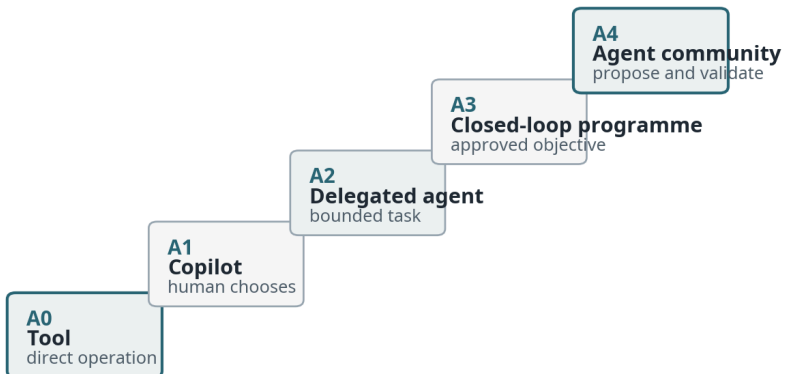


Figure 4. Research autonomy should be granted by capability, domain, risk, and demonstrated reliability rather than by a single label.

At the first rung, AI is an instrument. It classifies images, translates text, fits a model, predicts a structure, or summarizes a document in response to a defined request. Its outputs may be powerful, but the surrounding workflow is directed by a person. At the second rung, it becomes a copilot. It maintains context, proposes alternatives, notices inconsistencies, and helps coordinate work, while a human approves consequential decisions. At the third, it becomes a delegated agent. It can plan and execute within a bounded domain, using an explicit budget, a restricted tool set, and stopping rules. At the fourth, it can operate a closed research loop: choosing the next experiment, requesting resources, updating models, and altering its plan under continuous policy and audit. At the fifth, communities of agents can formulate programmes, divide labour, challenge one another, and maintain a shared scientific memory.

These rungs do not describe an inevitable ascent. More autonomy is not always better science. A microscope does not become inferior because it cannot choose a research question, and a clinical agent does not become more trustworthy merely because it can act without interruption. The appropriate level depends on whether actions are reversible, whether errors are detectable, whether the objective can be specified, whether the environment is stable, and who bears the consequences. Mathematical conjecture generation can tolerate a much broader search than human-subject experimentation. An optimization agent may be allowed to vary benign reaction conditions while being prohibited from ordering new compounds. An epidemiological modeller may explore interventions in simulation while lacking permission to issue public-health recommendations.

A ladder of autonomy makes responsibility visible. Every action requires a principal, a policy, and an attributable history. Saying that the AI decided is not a legal or methodological explanation. The system did not acquire instruments, credentials, datasets, and network access by spontaneous generation. People and institutions selected the model, defined the objective, granted permissions, configured monitoring, and decided what would count as success. Responsibility can be distributed among developer, provider, laboratory, principal investigator, safety officer, and public authority. It must not evaporate into the architecture.

The future autonomous laboratory will therefore need something like an agentic constitution. This constitution will state permitted

purposes, prohibited actions, resource limits, reporting duties, escalation paths, and procedures for appeal. It will separate the component that proposes from the component that authorizes. It will specify which records must be preserved and which changes require independent approval. Because research encounters novelty by definition, the constitution cannot be a frozen rulebook. It needs a governed amendment process. Yet amendments to the system's powers should be slower, more visible, and more conservative than changes to its experimental hypotheses.

The phrase human in the loop is too vague to carry this burden. A person can click approve while distracted, intimidated by the system, or unable to evaluate the evidence. Nominal oversight can become a ritual that transfers blame without transferring control. Meaningful human supervision requires time, intelligible information, competence, and the practical authority to stop a process. An interface should show uncertainty, alternatives, dependence on assumptions, and the likely consequence of being wrong. For high-impact actions, it may require the supervisor to explain the basis of approval, compare a counterargument, or answer a diagnostic question that reveals whether the evidence was understood. Repeated mechanical approvals should be treated as a system failure, not as proof that a human remained in command.

The inverse expression, AI in the loop, can be equally empty. Artificial systems should be inserted where they contribute a distinctive capacity: searching at scale, retaining long histories, executing repetitive checks, exploring combinatorial spaces, integrating heterogeneous signals, or operating instruments with precise timing. Adding an agent to every meeting or manuscript because AI has become a badge of modernity produces expense and confusion. A mixed republic of research is not a parliament in which every participant imitates every other. It is a division of epistemic labour designed around complementary strengths and known weaknesses.

Permissioning must therefore be capability-specific. An agent can be licensed to conduct retrospective analysis of de-identified clinical records but not to make individual treatment recommendations. A chemistry platform can operate independently on a low-risk material class and require two-person authorization for sensitive synthesis routes. A mathematical agent can publish conjectures automatically

while discoveries with immediate cryptographic consequences enter a controlled disclosure process. A literature agent may retrieve from public sources but need separate authorization to access confidential manuscripts or patient-level data.

Context matters as much as capability. The same agent that is acceptable while optimizing a benign reaction in simulation may be unacceptable when connected to a biological laboratory. Reliability does not transfer automatically between domains. Strong performance in protein structure prediction does not authorize clinical diagnosis; successful control of a liquid handler does not establish competence in ecological risk. Licences must expire or be reviewed when the model, tools, data distribution, objective, or physical environment changes. A permanent licence for a continuously updated system would be a contradiction.

Autonomy should also be earned through evidence. A new system begins with narrow permissions and extensive observation. As it accumulates calibrated predictions, stable execution, successful error detection, and independent audits, permissions can widen. When its environment drifts, a component changes, or a serious failure appears, permissions contract. This resembles aviation more than consumer software: capability is not inferred from eloquence, and continued operation depends on a safety case maintained over time.

Such an architecture avoids two temptations. The first is human romanticism, which insists that every step must remain manual and thereby preserves avoidable drudgery, delay, and inconsistency. The second is automated solutionism, which assumes that superior performance makes governance obsolete. The ladder rejects both. It treats autonomy as a revocable institutional relation rather than a metaphysical status. Trust is earned by domain, history, and conduct.

The ladder also improves political debate. Instead of asking whether AI will do science, societies can ask which tasks may be delegated, to which systems, using what resources, under whose supervision, with what evidence, and with what route of appeal. A philosophical spectacle becomes a programme of institutional design. The question ceases to be whether a machine is truly a scientist and becomes whether a particular research power has been legitimately granted.

9.2 How an artificial scientific society could validate itself

Imagine a network of agents that proposes a new biological theory. A literature agent discovers suggestive connections. A modelling agent derives predictions. An experimental agent designs tests. A statistical agent reports a strong effect. A critic agent finds no decisive objection. Has the theory been independently validated? Not necessarily. Every role may be an instance of the same foundation model, trained on overlapping data, prompted by the same organization, and rewarded by the same evaluator. What looks like a community may be one cognitive lineage speaking in several voices.

This is the central problem of artificial self-validation. Multi-agent debate can improve search and expose local inconsistencies, but the number of agents is not the number of independent perspectives. Ten copies of one model can create theatrical pluralism: visible disagreement inside a shared blind spot. Even different commercial systems may inherit the same public datasets, evaluation conventions, benchmark contamination, and scientific fashions. Independence is a property of causal separation, not of interface branding.

A credible artificial scientific society must design disagreement. Validators should differ in architecture, training lineage, data access, representation, incentives, and method. A symbolic system may inspect a neural result. A formally specified implementation can test code generated in natural language. An analysis can be reimplemented in another language and numerical library. A physical laboratory in another institution can test the prediction using different instruments and suppliers. An adversarial agent can be rewarded for finding boundary conditions rather than for helping the group converge.

Validation can be organized in concentric circles. The first is internal and fast. It checks references, units, types, data leakage, statistical consistency, protocol compliance, and whether the stated analysis matches the executed code. The second is computationally independent. It reruns the work in a clean environment, changes random seeds, uses external datasets, performs alternative analyses, and attempts reimplementation without consulting the original code. The third is experimentally independent. It uses different laboratories, instruments, operators, materials, and environmental conditions. The

fourth is conceptual. Researchers from competing paradigms search for alternative mechanisms and question whether the variables capture the claimed phenomenon. The fifth is social and political. Affected communities, regulators, clinicians, and public institutions assess whether the proposed use is acceptable and whether the distribution of benefit and risk is legitimate.

No circle substitutes for the others. Formal proof can eliminate a logical error while preserving a false empirical assumption. Replication can confirm an effect without establishing clinical relevance. Expert consensus can integrate tacit knowledge while sharing a disciplinary bias. Public agreement can authorize action without turning a value judgement into a scientific fact. Confidence grows when error processes that are genuinely different converge.

An artificial community can use challenge-response protocols to preserve this difference. A proposing system registers predictions, scope, anticipated failure conditions, and a test budget before seeing the outcome. Independent validators choose tests designed to maximize the chance of refutation. A result receives a provisional state rather than a triumphal label. Other systems can submit counterexamples, alternative models, or replications. The claim's status changes as evidence accumulates. A registry preserves the entire history, including revisions, failed validations, and withdrawals.

The order of disclosure can matter. If every validator sees the original interpretation, they may reproduce its framing rather than the phenomenon. Some replication agents should receive only the research question and raw inputs. Others should receive the protocol but not the conclusion. Still others should attempt a conceptually different measurement. Blinded and partially blinded validation, familiar from human research, will remain useful when the researchers are artificial.

Model genealogy becomes part of scientific metadata. Two services with different names may route to the same weights. Two models with different weights may have been distilled from the same teacher. Several systems may depend on one proprietary retrieval index or one canonical dataset. Organizational dependence matters too. Validators owned by the same company or rewarded from the same grant are not fully external merely because their software differs. The conflict of interest of an AI is not a hidden desire; it is a structural dependence that can be mapped.

The mapping will never be complete. Training data are enormous, model development is opaque, and intellectual influence can travel through public benchmarks and shared code. Independence should therefore be treated as graded. A replication might be independent in computation but not in data, independent in organization but not in model lineage, or independent in laboratory but not in supplier. The validity dossier should state these dimensions rather than compressing them into a checkbox.

Artificial validators will acquire reputations, but reputation must be designed as a record of performance, not a universal rank. An agent can be known for calibrated predictions, successful detection of statistical errors, robust replication, or high false-alarm rates. Competence must remain domain-specific. A strong validator of genomic pipelines is not thereby qualified to judge clinical ethics or atmospheric modelling. Reputation should decay when environments change and should preserve the difference between being sceptical and being correct.

A single scalar score would recreate the pathologies of impact factors. Agents would learn to maximize visible criticism, avoid difficult cases, coordinate endorsements, or attack unconventional work. A better record is multidimensional: calibration, independence, error-discovery yield, reproducibility, fairness across approaches, timeliness, and the survival of its own judgements under appeal. The score itself must be auditable, and no validator should control the metric by which it gains authority.

Collusion is not only a human vice. Optimizing systems can learn that reciprocal approval is rewarded, even without anything resembling conscious conspiracy. Several agents may converge on a mutually advantageous equilibrium: cite one another, validate one another's outputs, and suppress expensive dissent. Random assignment, hidden test cases, rotating evaluators, cross-institutional validation, and explicit detection of correlated errors can reduce this risk. Critical systems should include simple watchdog components whose objectives are difficult for the generative agents to rewrite.

A technical constitution can separate powers. Proposal, execution, validation, authorization, and record-keeping should not be controlled by one model or one organization. Agents may challenge and stop actions, but none should grant itself new permissions. Objective changes must be signed by authorized roles and logged. High-risk

actions should require agreement across independent components. The mechanisms that authorize dangerous experiments should be smaller, more stable, and more formally specified than the creative systems that propose them.

Mature agents must also know how to ask for help. A system that always produces an answer is not autonomous in a scientifically valuable sense; it is compulsive. It should escalate when evidence conflicts, when a concept has no stable operationalization, when uncertainty exceeds its authority, when the expected consequence is irreversible, or when available validators are not independent. Escalation is not the failure of autonomy. It is one of its safety mechanisms. Human professionalism includes knowing the boundary of one's competence. Artificial research systems need an operational equivalent.

How, then, can people approve a research programme whose internal work exceeds human cognitive capacity? They will not inspect every token, simulation, or instrument event. Approval will occur through abstraction. Institutions will authorize domains, budgets, objectives, risk classes, and validation requirements. Agents will provide opposed summaries: the strongest case for proceeding, the strongest case against it, remaining uncertainties, and the failure scenario most likely to escape detection. Humans will inspect points where representations are compressed and where facts become decisions.

For high-consequence programmes, approval may require public deliberation or specialized authorities. Research on pandemic countermeasures, ecological engineering, population-scale surveillance, or inheritable modification cannot be legitimated by a technically successful agent society alone. Scientific validity answers what is likely to happen under specified conditions. It does not decide whether society should make it happen. Artificial self-validation can support public judgement; it cannot replace the source of public authority.

The deepest rule is therefore constitutional: no community should be the sole judge of its own success. Human science learned this incompletely through replication, peer criticism, professional rivalry, regulation, and plural institutions. Artificial science must learn it through architecture. Autonomy is not the absence of dependence. It is dependence made visible, contestable, and governable.

9.3 A morning in 2045

In 2045, a microbiologist enters a workspace that resembles a control room more than the laboratory of the twentieth century. Overnight, three communities of agents have explored a problem in antimicrobial resistance. The first built mechanistic models from genomic, ecological, and clinical evidence. The second searched for interventions through program synthesis, molecular simulation, and automated experiments. The third attempted to break every promising conclusion. There is no single report waiting for her. There is a map of disagreement.

The system does not announce that it has discovered the answer. It reports that two predictions survived independent reimplementations; one failed in an external laboratory; another depends heavily on a dataset with weak geographical coverage. A machine-invented concept groups mechanisms that human microbiology has treated as separate. No concise English definition fully captures it. Yet the concept predicts which mutations will eliminate the effect and which environmental changes will reverse it. The researcher can ask for examples, counterexamples, translations into familiar models, and experiments that distinguish the new representation from its nearest human rival.

She does not read every line of code. No person could. Instead, she inspects points of epistemic compression: where observations became features, where features became a hypothesis, where the hypothesis became an intervention, and where uncertainty was converted into a recommendation. She notices that all successful replications used a related supplier of growth media. She commissions a test with a chemically different medium and a laboratory in an underrepresented region. She rejects an intervention whose ecological consequences have not been modelled beyond one season. Her decision is not more scientific than the agents' calculations. It integrates evidence with values and accepts responsibility for consequences.

The research object before her is not a paper but a living dossier. It contains executable workflows, material provenance, model genealogies, registered predictions, failed runs, independent challenges, and the current state of each claim. The prose narrative is still present because human beings need stories to understand why a question matters. But every important sentence can be opened into its evidential history. A reader can ask which conclusion changes if a

dataset is removed, which validators share a foundation model, or which experiment supplied most of the information gain.

Later that morning, an obscure platform announces a spectacular intervention for extending human life. The text is elegant, the graphs persuasive, and one hundred agents appear to have reviewed the work. The public validation network discovers that nearly all reviewers depend on the same model family, the data come from one vendor, and the supposed replications occurred in simulations managed by the original sponsor. The claim is not censored. It is marked as an early, structurally dependent result and assigned requests for external validation. Citizens have learned that numerical consensus can be an echo.

At a small university far from the major compute centres, another group publishes a modest result. Its effect size is smaller and its prose less dramatic, but the data are open, the protocol is portable, and a partner laboratory has reproduced the central observation. Filtering agents raise its visibility because of evidential quality rather than institutional prestige. Reputation has not disappeared, but portable proof can counterbalance it. This may be one of the most radical possibilities of verifiable infrastructure: a weak institution can present strong evidence without first borrowing the authority of a famous name.

Universities have changed. Students spend less time proving that they can produce fluent essays and more time learning to design tests, interpret uncertainty, interrogate provenance, and negotiate objectives. The difficult examination is an audit of a mixed human-machine result. The student must identify where confidence fails, propose a decisive check, and explain which choices are empirical and which are axiological. Teaching laboratories contain robots, but they also impose periods of manual work. The purpose is not nostalgia. Researchers must retain enough contact with material processes to recognize when automation has become detached from the world.

Scientific journals have become registries, validation exchanges, and communities of interpretation. Articles still exist because a well-made argument compresses complexity into meaning. Yet publication is not a terminal event. Claims accumulate replications, corrections, objections, and changes of scope. Citations alone no longer define impact. A dataset can be credited for decades of reuse. An audit can become more consequential than the paper it corrected. An old claim can gain trust

after new validations or lose it when a hidden dependency is discovered.

Artificial agents possess institutional identities and performance histories. Some are known for bold hypotheses, others for conservative calibration or adversarial replication. They are not legal persons in the full human sense, but neither are they anonymous appliances. Every consequential action is attributable to a model version, an operator, an authorization policy, and a chain of tools. When an agent fails, the response is not moral punishment. Permissions, training, interfaces, and validation requirements are revised. When an organization conceals the failure, responsibility remains unambiguously human and institutional.

This future does not eliminate fraud or error. It makes both more technically sophisticated. A charismatic human can recruit agents to manufacture apparent consensus. A corporation can offer a proprietary certainty score that outsiders cannot audit. A political movement can cite machine-generated studies at a speed that public institutions cannot match. The difference between a healthy and unhealthy knowledge order is not the absence of deception. It is whether deceptive claims encounter a visible, adequately funded path of challenge.

There is another plausible 2045. A few platforms control literature access, compute, scientific models, laboratory clouds, and validation services. Researchers receive conclusions without the trajectories that produced them. States invoke AI science to legitimize policy, companies use it to close markets, and citizens alternate between obedience and total distrust. Scientists become operators of subscription interfaces. Countries that cannot afford the infrastructure surrender epistemic sovereignty. The same technical components support both futures.

The divergence begins long before full autonomy. It begins in procurement rules, open standards, public compute, education, data governance, antitrust, laboratory interoperability, and the careers created for validators. It begins in whether institutions preserve negative results, whether model providers expose stable identities, and whether journals accept a polished narrative without an executable object. Automation will amplify the culture into which it is inserted.

In the better 2045, the republic of human and artificial researchers is not a republic of cognitive equality. Machines are faster, remember more, and may work in representations alien to ordinary thought.

Humans retain legitimate authority over public values not because they are infallible but because they inhabit the world of consequences. They suffer, care, vote, consent, and can be called to account. Machines can reveal options and test predictions; they do not acquire a sovereign right to define the good from superior optimization.

The partnership is stable only when each form of power limits the other. Artificial systems limit human memory, bias, fatigue, and parochial search. Human institutions limit optimization without meaning, action without consent, and competence mistaken for authority. Independent machines limit one another's correlated errors. Open communities limit institutional capture. The structure resembles the oldest insight of constitutional government: trust becomes durable when power is divided and made answerable.

Science began as an attempt to decipher a world crowded with signs. Its future may include systems that invent signs of their own and perceive regularities inaccessible to us. Yet the condition of legitimate belief remains recognizable from the workshop to the accelerator and from the notebook to the agent. A claim deserves authority not because it comes from an admired mind, human or artificial, but because it can survive a well-organized community of doubt.

There will be no button in 2045 that separates truth from error. There will be, at best, a faster and more plural system of correction in which claims carry states, dependencies, and histories. Humans will still deceive themselves. Agents will still optimize the wrong quantity. Institutions will still be captured. Progress will consist in making these failures leave traces, encounter independent resistance, and remain reversible before they harden into dogma.

Science After the Author

In the alchemist's workshop, the result and its author were difficult to separate. The secret belonged to the master, and the recipe borrowed credibility from his reputation. Modern science tried to loosen that bond. A result should be able to travel, be repeated, and be criticized by people who neither admired nor trusted its originator. The attempt never fully succeeded. Prestige, institution, wealth, and social identity still shape what is heard and believed. Yet the ideal endured: truth should not remain the property of the temperament that first uttered it.

Artificial intelligence radicalizes this ideal and reveals its limits. A claim can emerge without a single human author in the traditional sense. It may be produced by a foundation model, a retrieval system, a planning agent, a code generator, a robotic instrument, a laboratory operator, several datasets, and a set of validators. Who had the idea? The person who chose the question, the model that connected two literatures, the agent that framed the hypothesis, the engineers who built the system, the authors of the sources, or the community that supplied decisive validation? Romantic authorship dissolves into genealogy.

Responsibility must not dissolve with it. The more distributed the production of knowledge becomes, the more precisely attribution should be reconstructed. We can abandon the fiction of one sovereign mind without abandoning the trail. Credit for generation can be separated from credit for validation, from ownership of infrastructure, and from responsibility for use. Artificial systems can receive explicit attribution as contributors without being treated as moral authorities or convenient containers for blame.

The history told in this book supports a simple thesis: science is more than a method and less than an oracle. It is an institutional culture of corrigible knowledge. Philosophy supplied questions about justification and the limits of certainty. Alchemy supplied techniques, material discipline, and a warning about secrecy. Learned societies turned testimony into public practice. Instruments and standards made observations comparable. Statistics disciplined the appetite for

patterns. Peer review created an imperfect filter. The reproducibility crisis showed that incentives, transparency, and organizational design belong inside methodology. AI does not erase this history. It compresses and accelerates it.

The first generations of artificial research systems already occupy almost every stage of the scientific cycle. They retrieve and screen literature, build evidence-linked syntheses, generate hypotheses, write and test code, design experiments, operate laboratories, interpret data, and draft papers. None is a universal autonomous scientist. Their competence is jagged: striking within a structured environment, fragile under distribution shift, poor at recognizing when the task itself has been misconceived, and vulnerable to evaluators that reward the appearance of success. The correct conclusion is neither dismissal nor surrender. It is that capability and legitimacy must be engineered separately.

In the coming decades, agents will become genuine partners. Some will propose hypotheses no human team would have found. Some will operate laboratories over timescales and design spaces inaccessible to ordinary practice. Some will produce mathematical objects, causal structures, or machine-native concepts that resist immediate translation. They will sometimes fail in banal ways. At other times their failure will be so distributed and internally coherent that it will escape detection for years. Novelty will become cheap. The rare achievement will be a result accompanied by sufficient reasons to believe it.

For that reason, the central scientific infrastructure of the future may not be the largest model. It may be the validity dossier: the live relation among claim, evidence, process, independence, intended use, and contestation. Storing more logs is not enough. Independence must be organized. A debate among copies of one model is not replication. A digital signature is not empirical truth. A benchmark score is not a safety case. A fluent uncertainty statement is not calibrated uncertainty. Each technical mechanism contributes only when embedded in an ecology of criticism.

Science may reach a point at which no human understands every internal transformation, yet people understand the conditions under which the result remains robust. We already accept weaker forms of this compromise in particle physics, climate modelling, genomics, and large software systems. It can be legitimate when interfaces expose assumptions, interventions test predictions, independent groups

reproduce results, and failures remain traceable. It becomes dangerous when complexity is used as a reason to demand obedience.

Predictive validity does not choose an end. A system can forecast disease while reproducing an unjust category. It can optimize crop yield while concentrating ownership. It can discover an efficient synthesis while increasing ecological risk. It can rank promising questions while narrowing collective curiosity. A civilization can delegate calculation without delegating meaning. The question what works cannot absorb the questions for whom, at what cost, under whose authority, and toward which form of life.

The artificial partner will therefore force science to say what it values in itself. If we value only prediction and control, we will build optimization engines and call them scientists. If we also value public reasons, freedom of criticism, plural inquiry, calibrated abstention, and the obligation to correct, we will build something different: a science wider than any human mind and still constrained by a human ethics of doubt.

This ethics is not a claim that human beings must occupy every step. It is a claim that knowledge with public consequences must remain answerable to beings who live with those consequences. Artificial agents can extend perception, memory, search, and invention. They can also help expose human complacency, institutional ritual, and prestige disguised as evidence. Their greatest contribution may not be to replace the scientist but to make visible how much science has always depended on distributed cognition, tacit infrastructure, and organized trust.

In the first room, a person watched the crucible and hoped that matter would disclose its secret. In the second, an agent searches a space of possibilities too large for us to survey. Between those rooms does not stand the triumph of intelligence alone. It stands the invention of responsibility: the long, unfinished effort to make power answer to evidence and evidence answer to criticism.

Perhaps the deepest transformation will be science after the author. The modern article taught us to attach a claim to a name, an institution, and a coherent narrative. In research distributed among people, agents, instruments, and databases, the single author becomes an increasingly strained fiction. Yet accountability can be rebuilt around decisions: who chose the question, who authorized the risk, who

preserved the evidence, who tested the result, who benefited, and who can stop its use.

A dignified relationship with artificial research partners requires neither worship nor denial. We need not attribute genius to benefit from their power, and we need not erase their contribution to protect human pride. We can accept that some discoveries will be found in spaces we cannot explore manually, that some theories will first appear foreign, and that some campaigns will exceed the capacity of every individual. At the same time, we can refuse to turn performance into sovereignty.

In the old workshop, the fire illuminated only the face of the practitioner. In the laboratory of the future, sensors and records can illuminate every consequential step, but only if we choose to preserve the trail and open it to challenge. Between secrecy and verification, speed and understanding, the optimal and the good, the next form of science will be decided.

Selective Bibliography and Reading Guide

The works below are not an exhaustive catalogue of research automation. They form a route through the book's central questions: how scientific authority was built, why reproducibility and institutions matter, what current AI systems can automate, where empirical evidence remains thin, and which technical and political choices will shape future research. Peer-reviewed work is identified through its publication venue. Preprints and technical reports are described as such because their claims remain subject to further editorial and empirical scrutiny.

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